



Naveed Ishaque
SPODA 2026, Uppsala



Cell-segmentation-free analysis



IshaqueLab

Bioinformatics to understand biological heterogeneity

AG-Ishaque



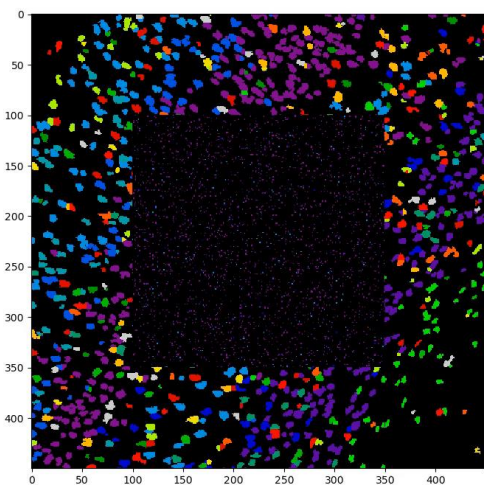
Postdoc



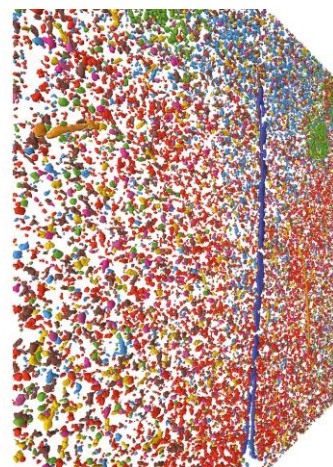
PhD



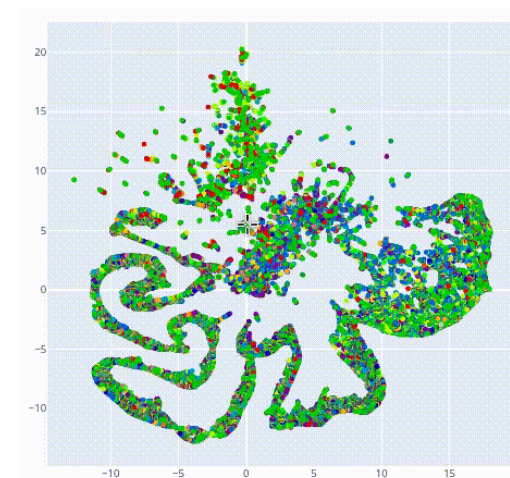
Clinician Scientist



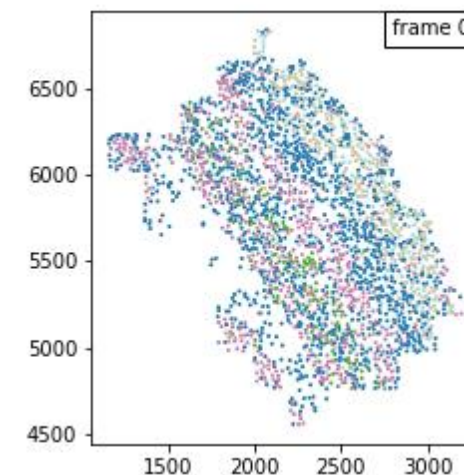
In-silico tissue generation



3D reconstruction



Interactive analysis tools



Domain identification

Learning objectives

Understand that accurate cell segmentation remains a problem

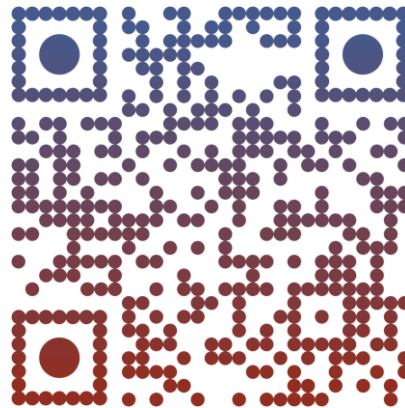
- Cell-segmentation-free analysis is an attractive alternative

Practical:

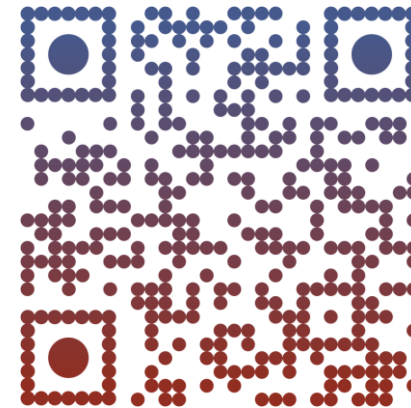
- Using the Saisnc tool to perform cell-segmentation-free analysis
- Trainers: Dr. Aditya Singh (NBIS, SE) and Dr. Lucie Pfeiferova (CAS, CZ)



Tutorial



Saisnc GitHub



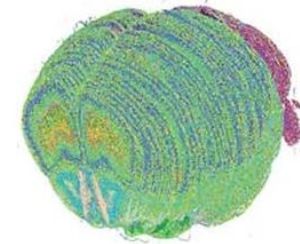
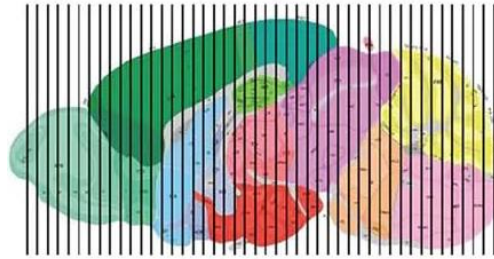
Saisnc Docu

Emerging state of the art for spatial transcriptomics

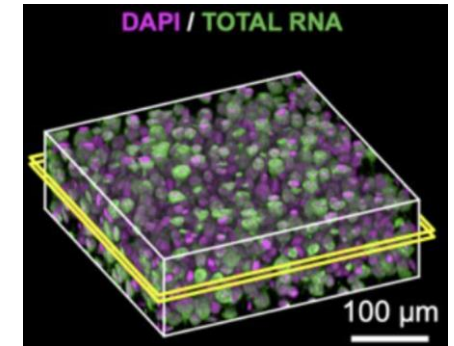
Size and resolution



Pseudo 3D

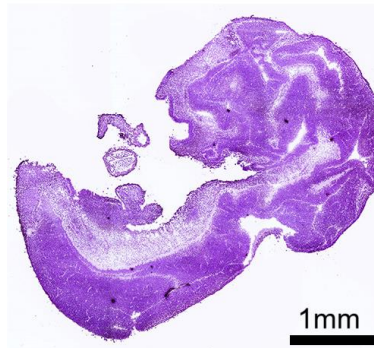


Real 3D



Multi-omics

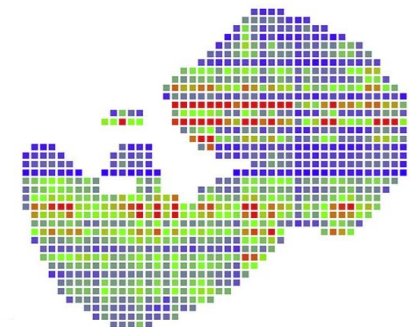
H&E (E.10 Embryo)



Pan-mRNA map



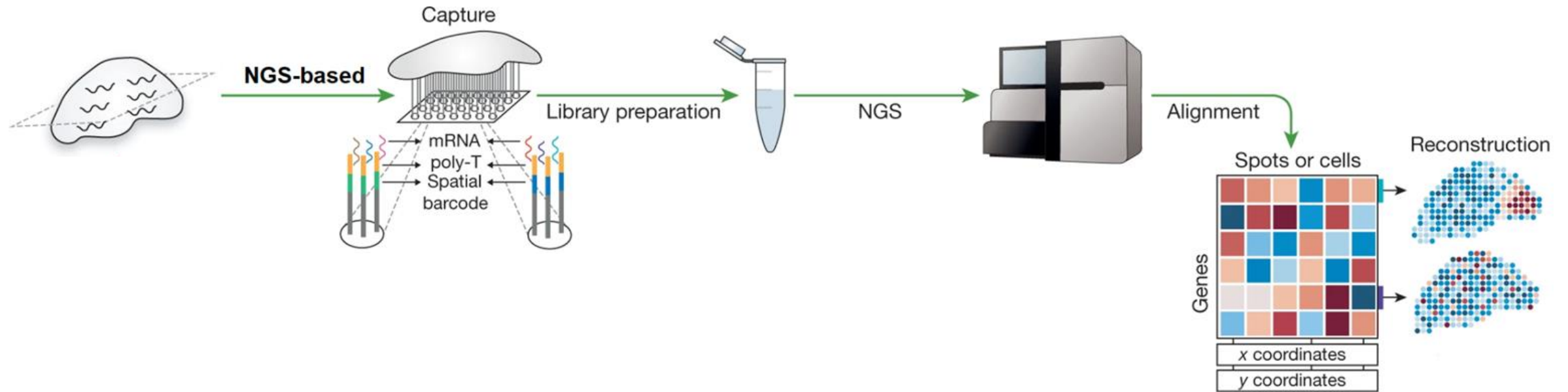
Pan-protein map



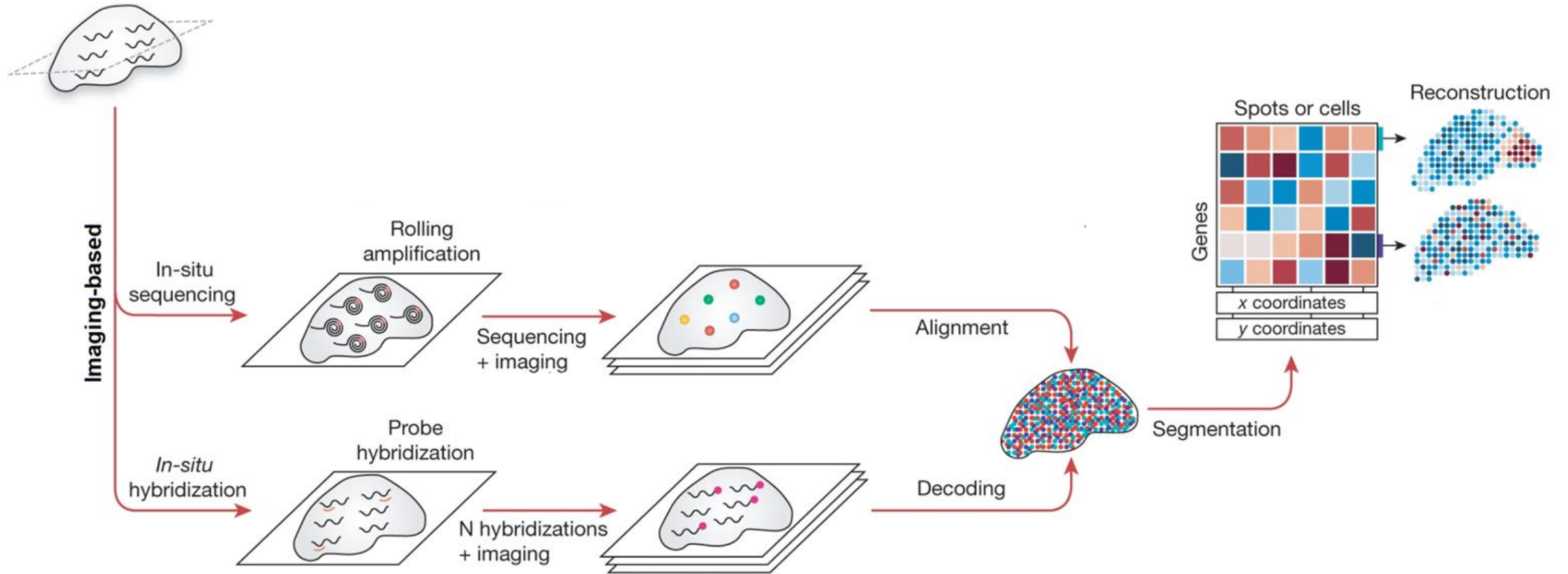
Spatial transcriptomics methodologies



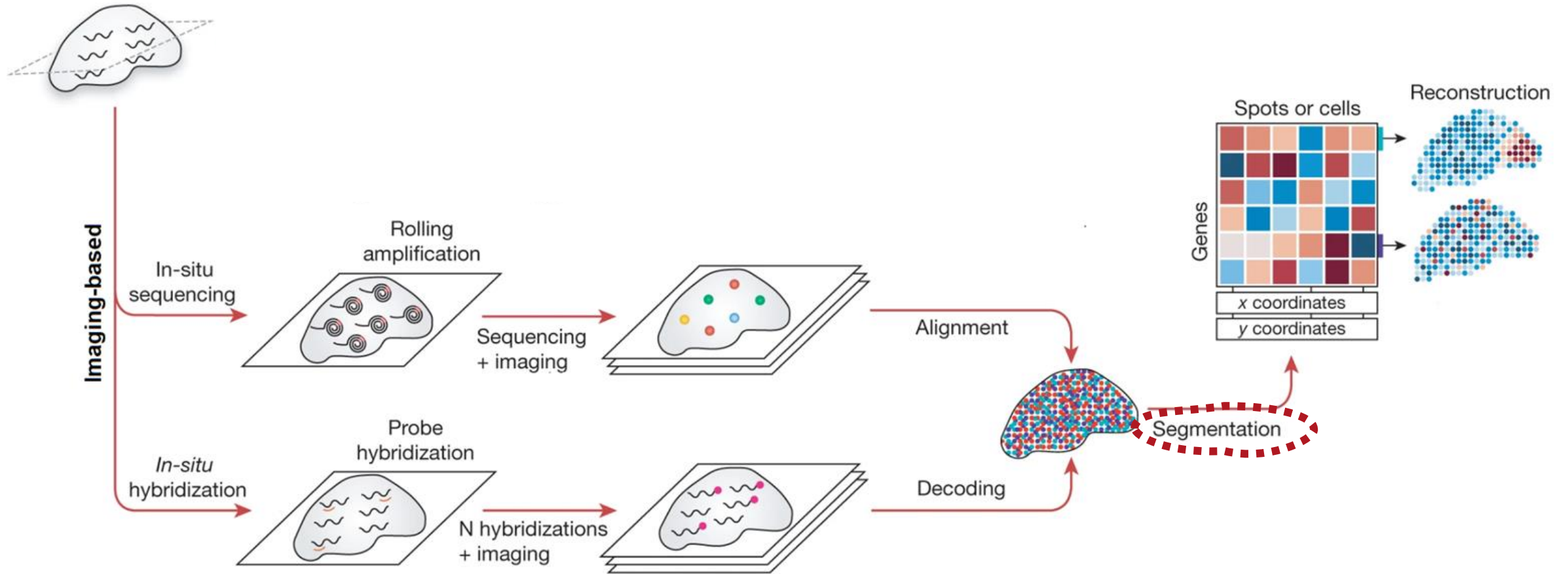
Spatial transcriptomics methodologies: NGS-based



Spatial transcriptomics methodologies: imaging-based

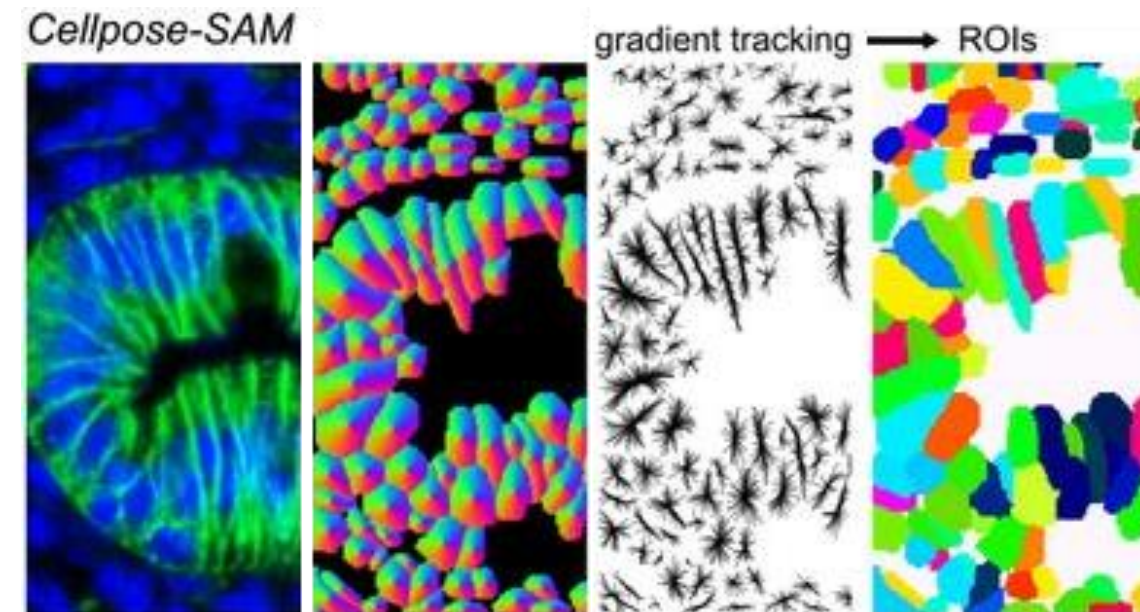
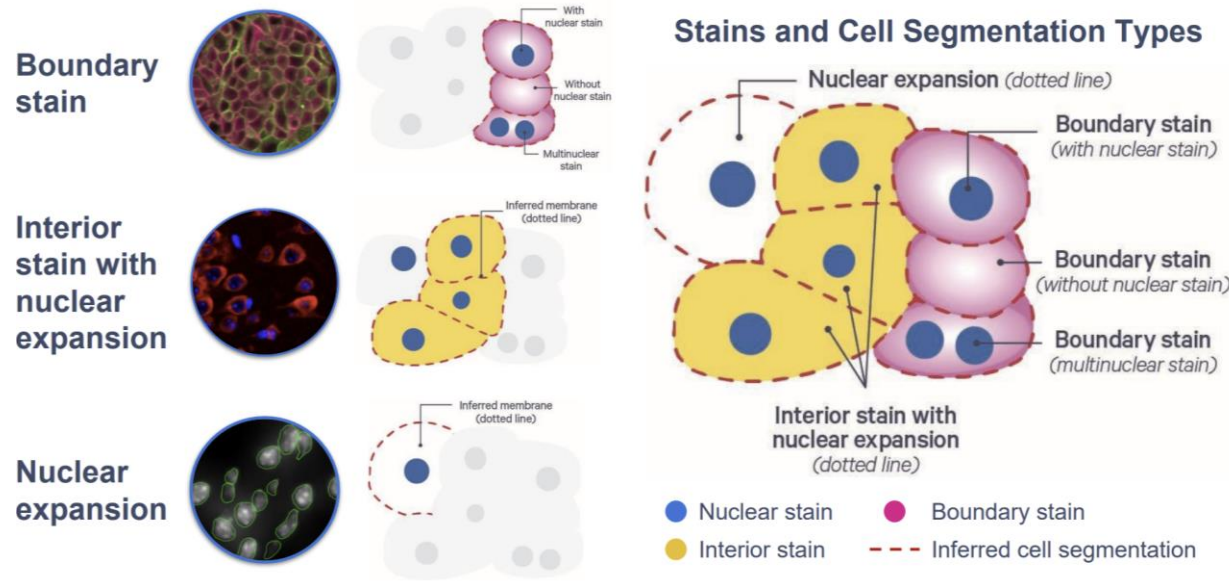


Spatial transcriptomics methodologies: imaging-based

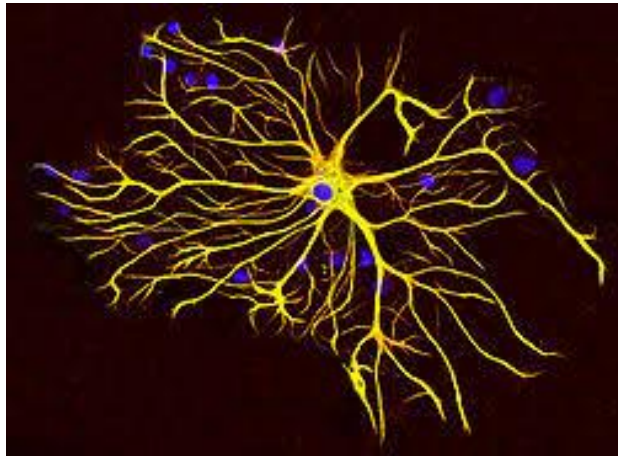


Recap: state-of-the-art of cell segmentation strategies

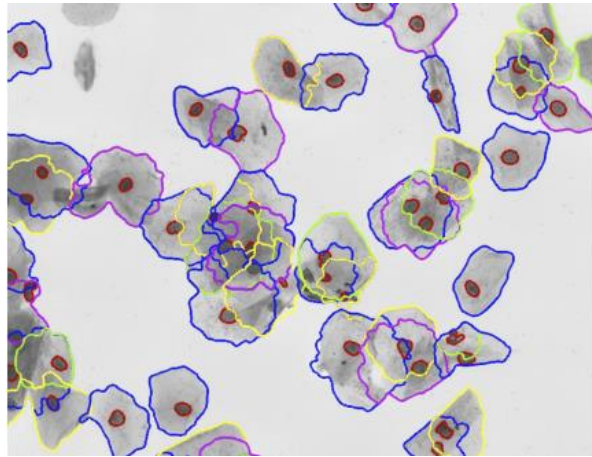
- Staining of multiple cell landmarks for “multi-modal” cell segmentation (e.g. 10x)
- Machine learning image analysis computational tools (e.g. CellPose-SAM)



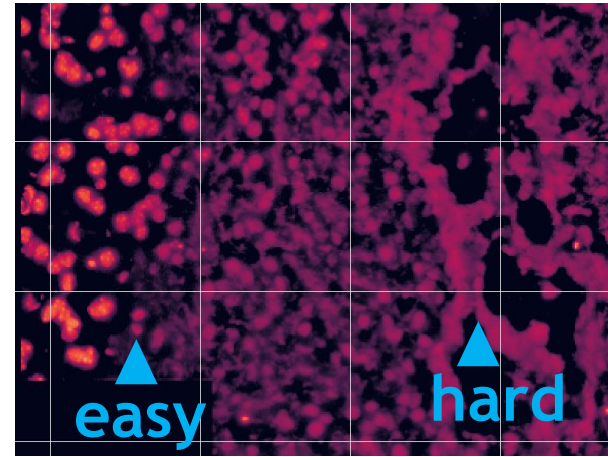
Cell segmentation is challenging



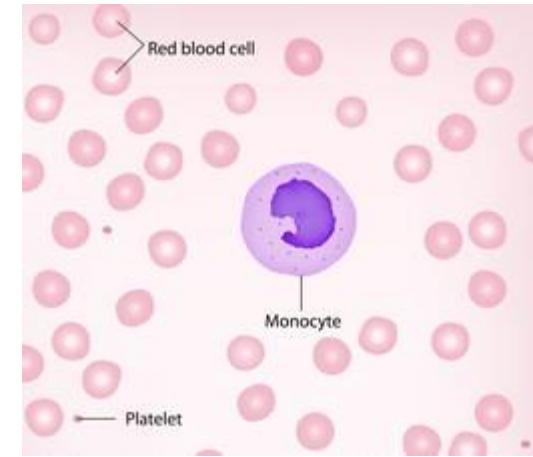
Complex shapes



Overlapping cells



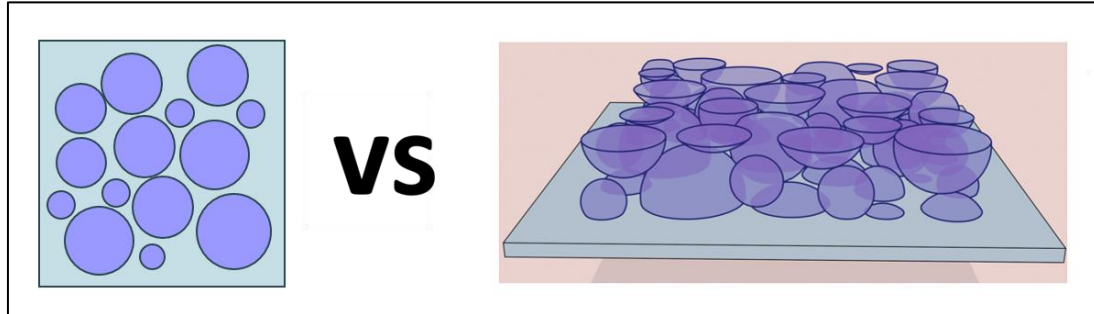
Cell stains might have issues



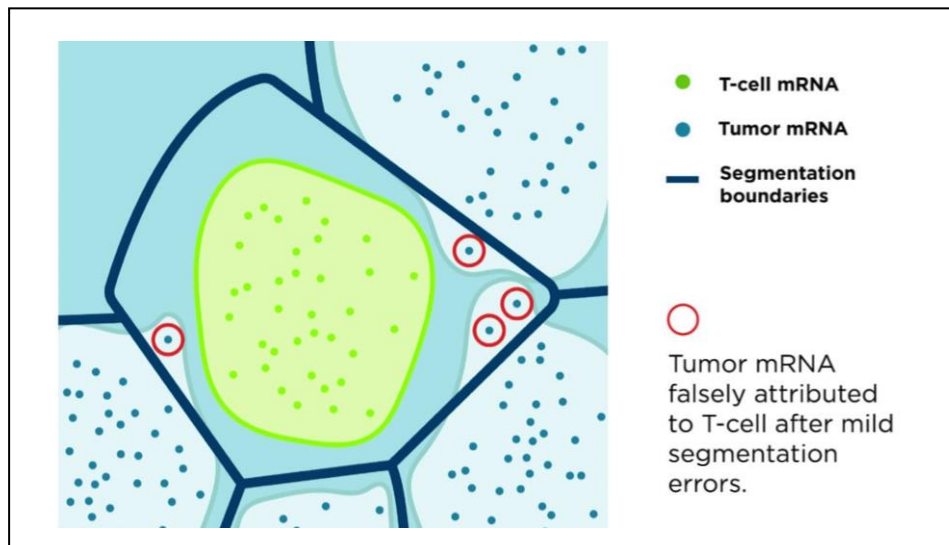
Cells might not be stained correctly (e.g. red blood cells)

- Performing cell segmentation is challenging
- We need more efficient ways to get first insights into data
- Cell-segmentation-free approaches are fast, scalable, and provide insights into the data

Perspectives on cell segmentation quality control



Tissue sections are 3D and not 2D. Cell overlaps are a problem!



Transcripts from adjacent cells can be miss-assigned!

arXiv > q-bio > arXiv:2606.09675

Quantitative Biology > Other Quantitative Biology

[Submitted on 8 Jun 2026]

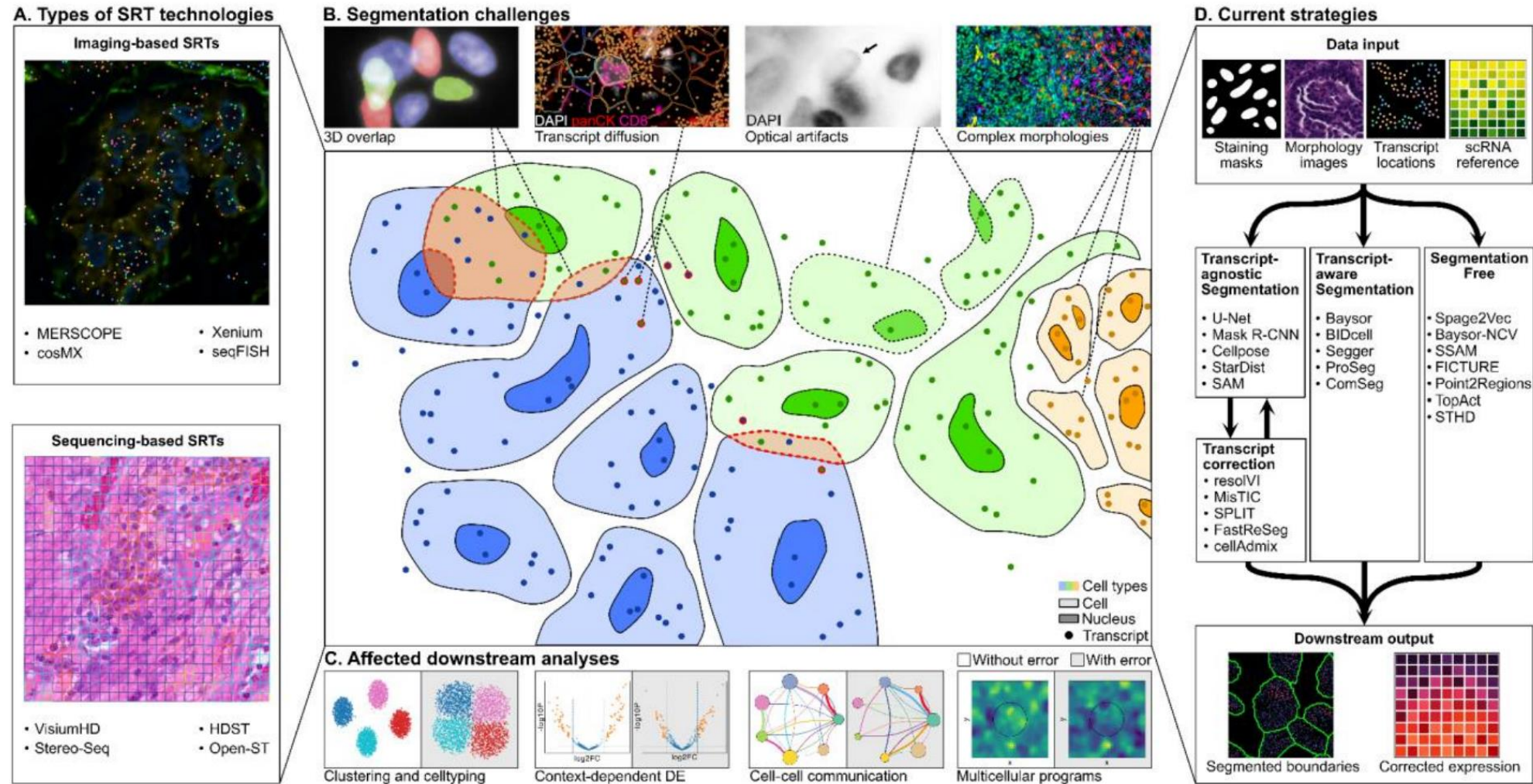
The Challenge of Cell Segmentation in Spatially Resolved Transcriptomics

Naveed Ishaque, Peter Kharchenko, Daria Lazic, Jieran Sun, Jean Yee Hwa Yang, Martin Emons, Florian Heyl, Wolfgang Huber, Daniel Jones, Louis B. Kuemmerle, Alex R. Lederer, Malte D. Luecken, Vinicius Maracaja-Coutinho, Matthias Meyer-Bender, Andrew Moorman, Evan W. Newell, Quan Nguyen, Shyam Prabhakar, John Randell, Daria Romanovskaia, Oliver Stegle, Gary D. Bader, Raphael Gottardo

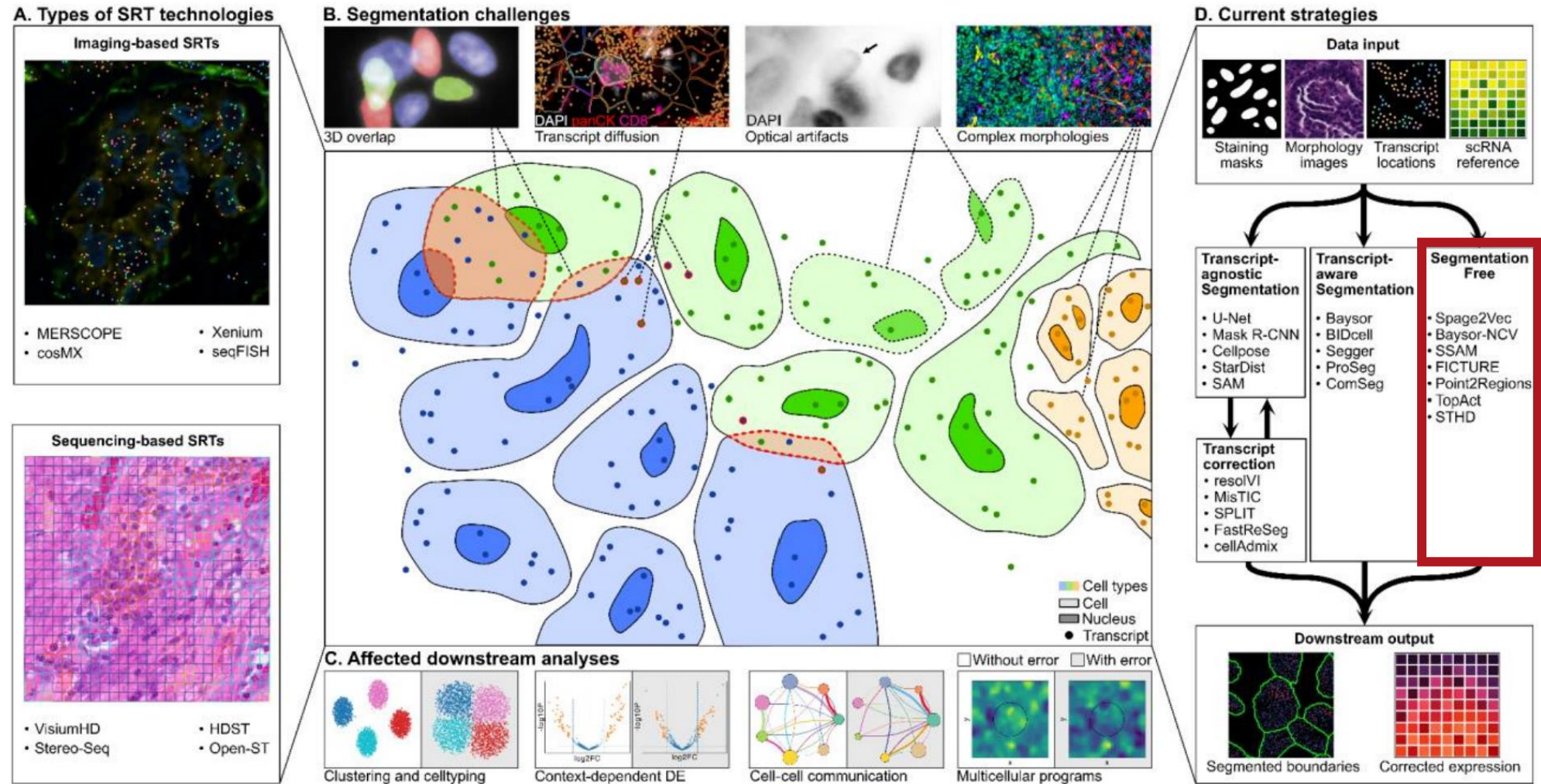
Spatially resolved transcriptomics (SRT) is transforming how we study tissues by measuring gene expression in cells in their spatial context. However, the field lacks robust methodological guidance on one of its most fundamental analytical steps: how to accurately segment cells and assign spatially localized transcripts to them. Major technical challenges include sparse molecular signals, transcript displacement, complex cellular morphologies, and the projection of three-dimensional tissue architecture onto two-dimensional imaging planes. These challenges make segmentation a major source of uncertainty, with errors that can propagate through downstream analyses and ultimately lead to misleading biological interpretations. Here, we argue that segmentation should be treated as a central unresolved problem in spatial omics rather than a routine preprocessing step. We review current approaches, highlight key methodological limitations, including the lack of appropriate metrics and gold-standard benchmarks, and propose a community-driven path forward. Establishing shared evaluation frameworks, scalable benchmark datasets, and transparent reporting standards will be essential for transforming SRT into a robust and reproducible foundation for biological discovery and clinical translation.

Perspectives on where things stand, the challenges, and what's next

Challenges and downstream effects of bad segmentation

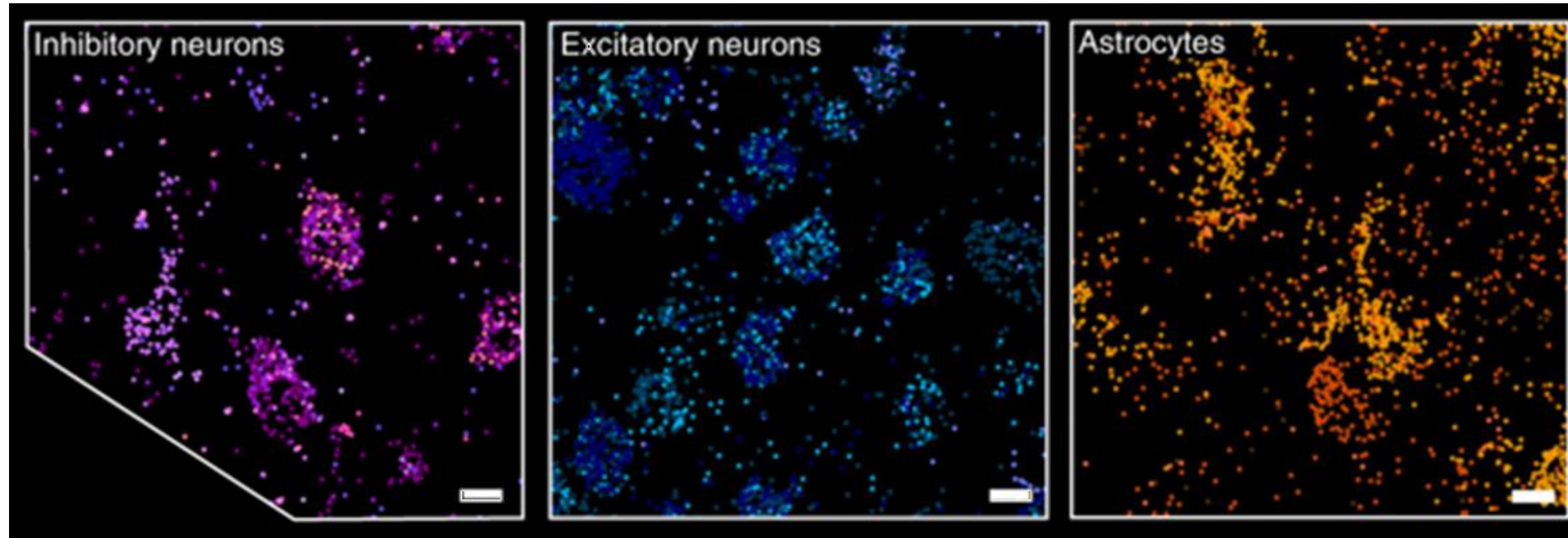


Challenges and downstream effects of bad segmentation

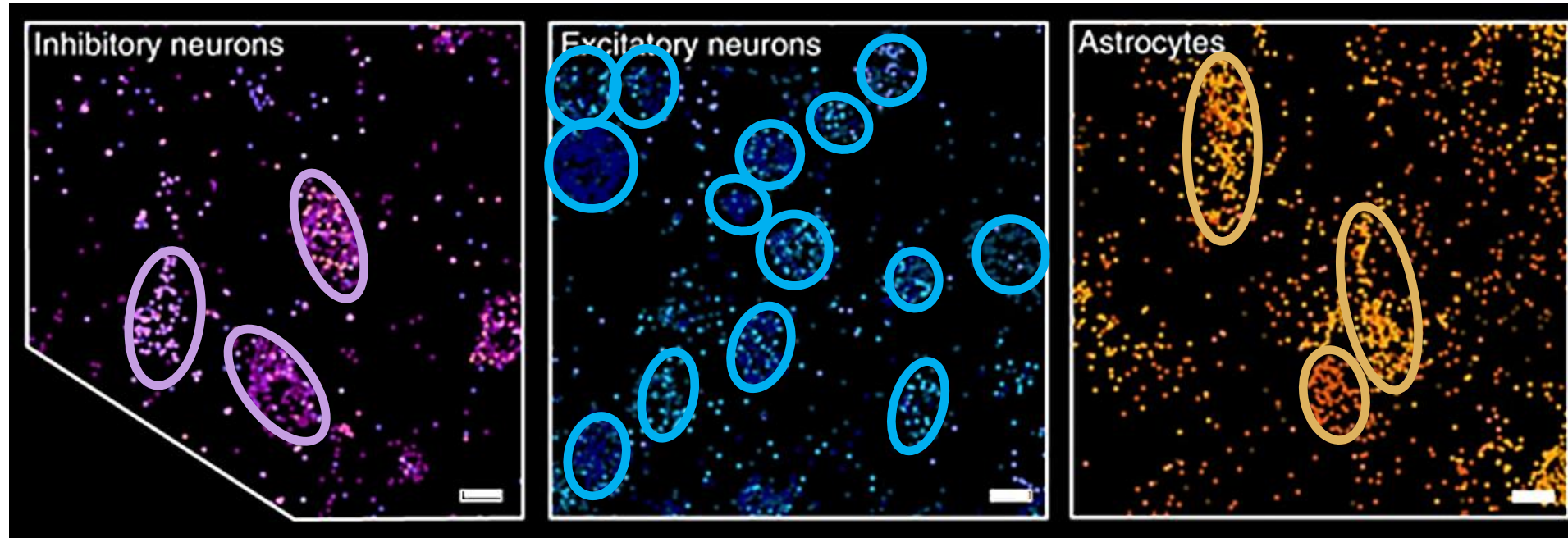


Cell-segmentation-free analysis

Transcript organisation patterns are not random



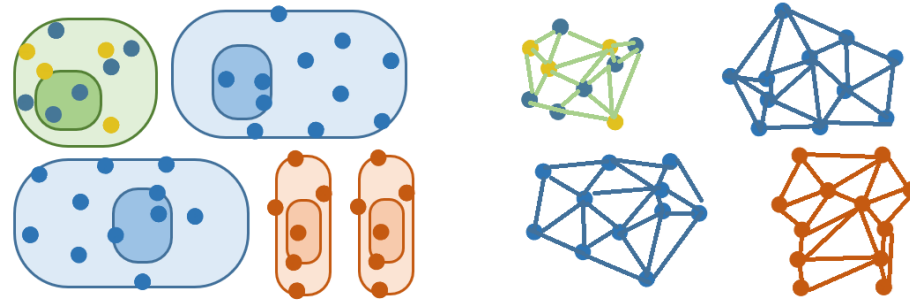
Transcript organisation patterns are not random



Modelling transcript distribution

Spatial model - how are mRNA molecules organised when they come from the same cell?

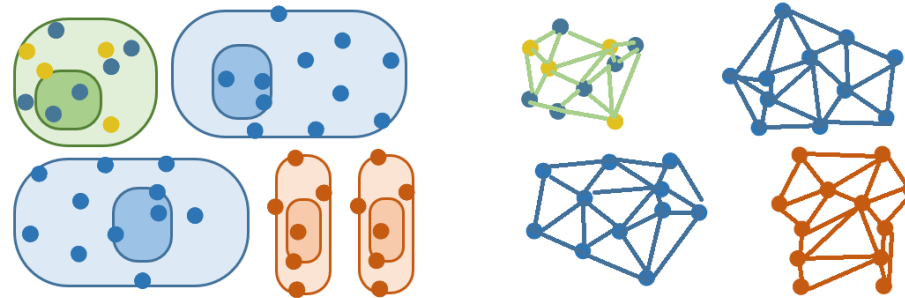
- Graph-based models



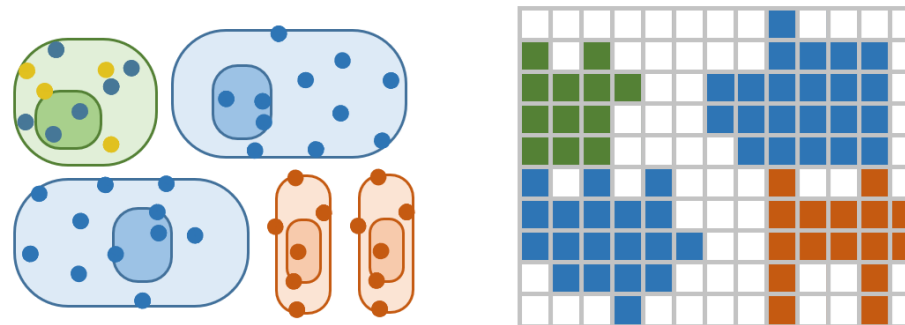
Modelling transcript distribution

Spatial model - how are mRNA molecules organised when they come from the same cell?

- Graph-based models



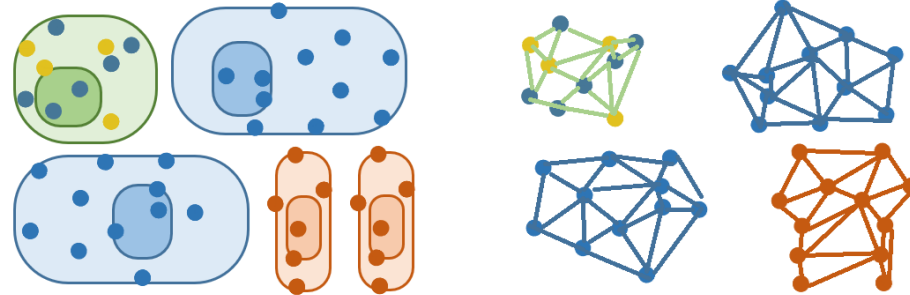
- Density-based models



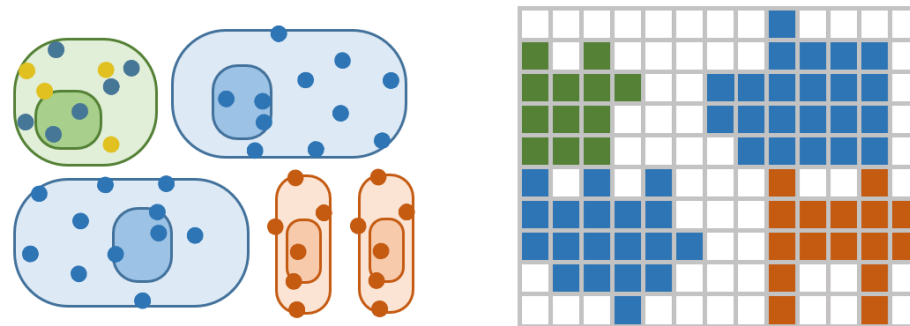
Modelling transcript distribution

Spatial model - how are mRNA molecules organised when they come from the same cell?

- Graph-based models



- Density-based models



Cell type model - how is spatial gene expression associated to different cell types?

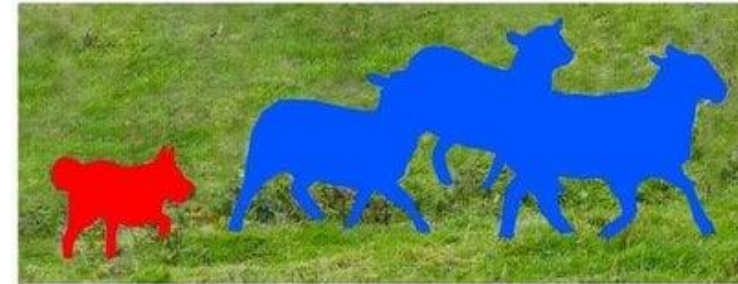
- Prior cell type specific expression signatures e.g. from single cell RNA sequencing data

Cell-segmentation-free... is still segmentation

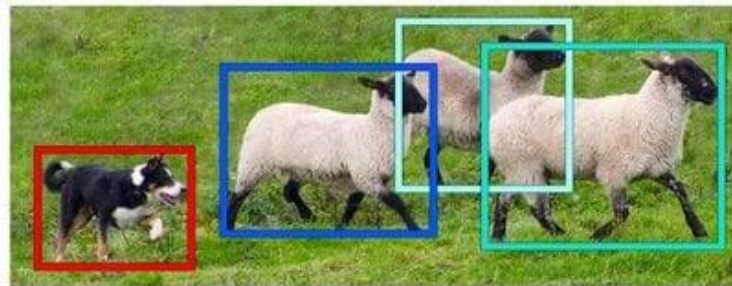
- Cell segmentation is instance segmentation (separate cells)
- Cell-segmentation-free can be semantic segmentation (labelled pixels)
- Cell-segmentation-free can be instance segmentation (e.g. not cells, but maybe cell parts)



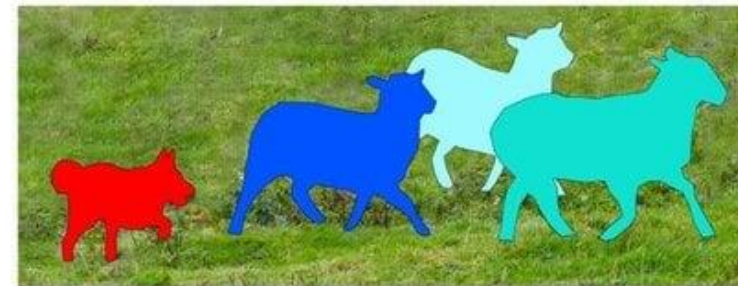
Image Recognition



Semantic Segmentation



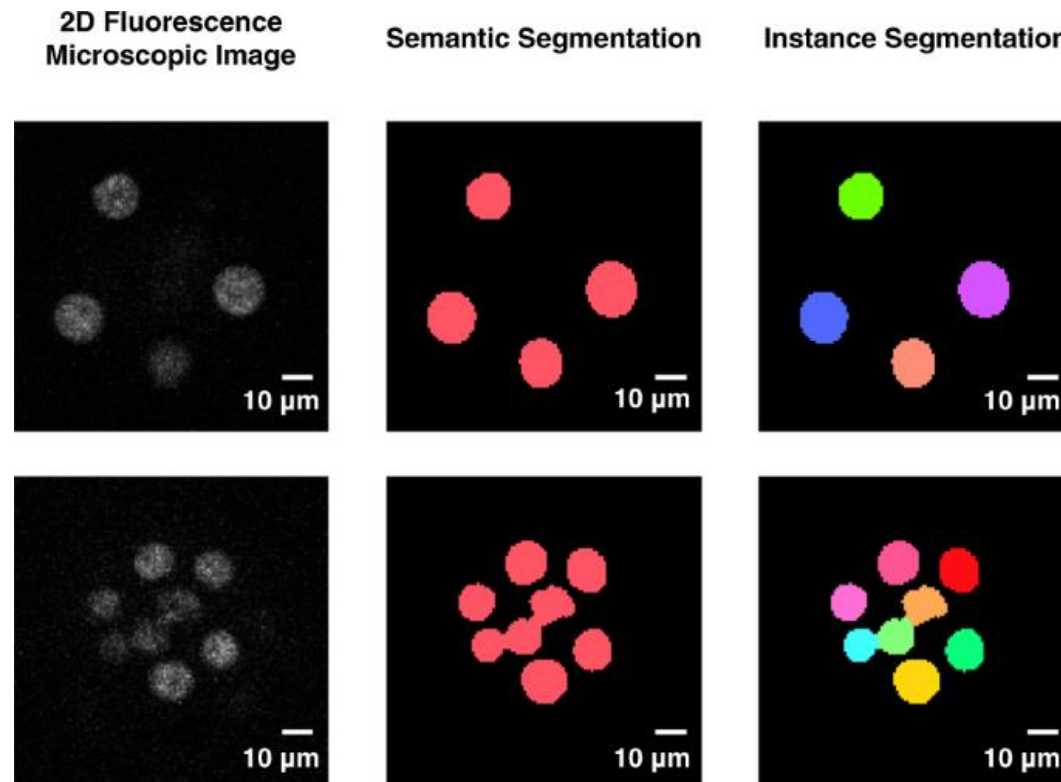
Object Detection



Instance Segmentation

Cell-segmentation-free... is still segmentation

- Cell segmentation is instance segmentation (separate cells)
- Cell-segmentation-free can be semantic segmentation (labelled pixels)
- Cell-segmentation-free can be instance segmentation (e.g. not cells, but maybe cell parts)



Cell-segmentation-free: analysis tools

- **Graph-based models**
 - spage2vec (Partel and Wählby, FEBS J, 2020) ← created right here in Uppsala!
 - Baysor-NCV (Petukhov et al, Nat Biotechnol, 2021)
 - Points2Regions (Andersson et al, Cytometry A, 2024)
- **Density-based or binning models**
 - SSAM (Park et al, Nat Commun, 2021) and SSAM-lite (Tiesmeyer et al, Front Genet, 2022) *
 - TopACT (Benjamin et al, Nature 2024)
 - **SAINSC (Mueller-Boetticher et al, Small Methods, 2025) ***
 - FICTURE (Si et al, Nat Methods, 2024)
 - STHD (Sun et al, Genome Biology 2025)
- **What about tools like Baysor and Proseg?**
 - They are not included as they perform instance segmentation, with the objective to identify cells
 - They are the first cell segmentation tools, that do not necessarily rely on imaging data

Cell-segmentation-free: major pros and cons

Pros:

- Generally, requires less computational resources
- Not limited to stains (e.g. red blood cells have no nucleus, so DAPI isn't useful)
- Rather than focussing on images, the focus is on transcripts (e.g. the primary analyte)

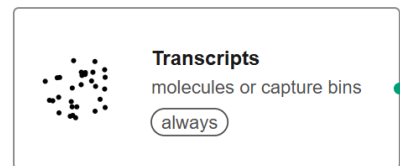
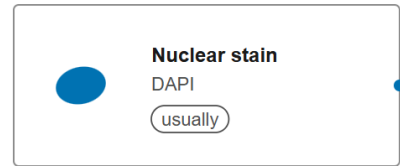
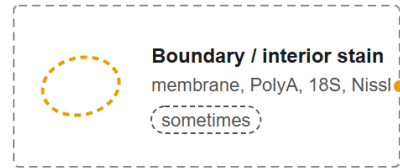
Cons:

- Cannot identify cells without transcripts (e.g. if a cell-type marker didn't work)
- Conceptual interpretation of results - where are my cells?
- Limited downstream analysis options

Comparative overview of cell-segmentation-free

Data (SRT readout)

solid = routine, dashed = often absent

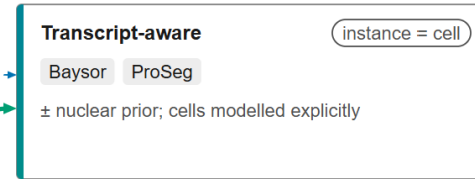
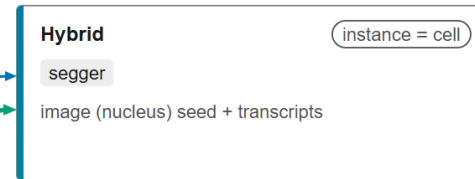
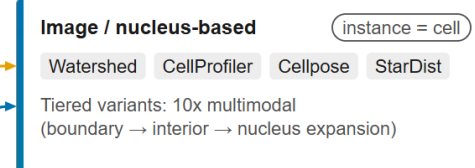


Connectors

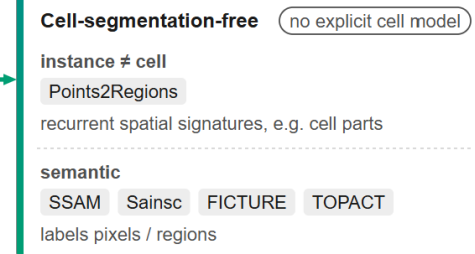
- transcripts
- nuclear stain
- boundary / interior stain
- optional input
- method → output

Segmentation approach

ordered image-driven → transcript-driven

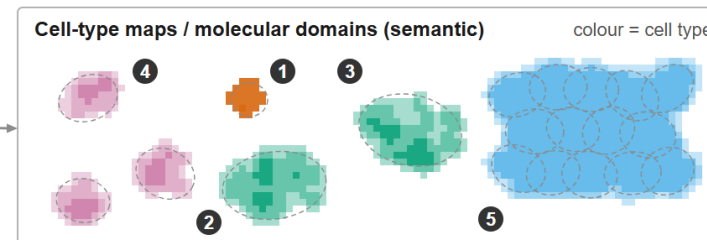
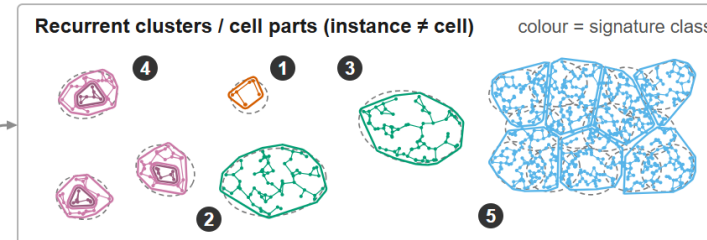
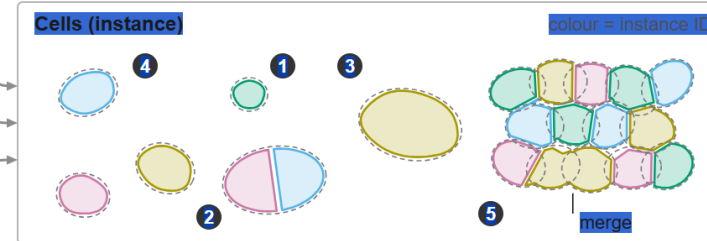
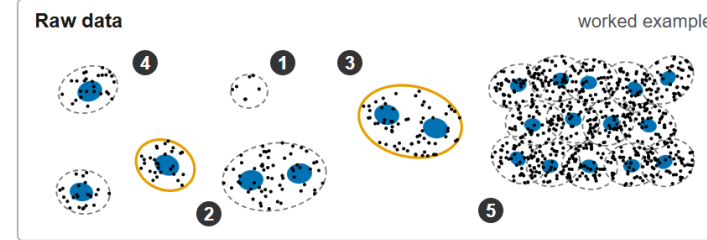


cell segmentation (explicit cell model)
cell-segmentation-free



Output

one tissue field, segmented three ways



Panel key

- nucleus (DAPI), measured
- membrane stain, measured where present
- true cell extent (not stained)
- ⋯ transcripts (fill full cell extent)
- segmented cell (solid outline + tint)
- ⬢ recurrent cluster (graph + outline)
- semantic pixels (pixel fill)

Worked example

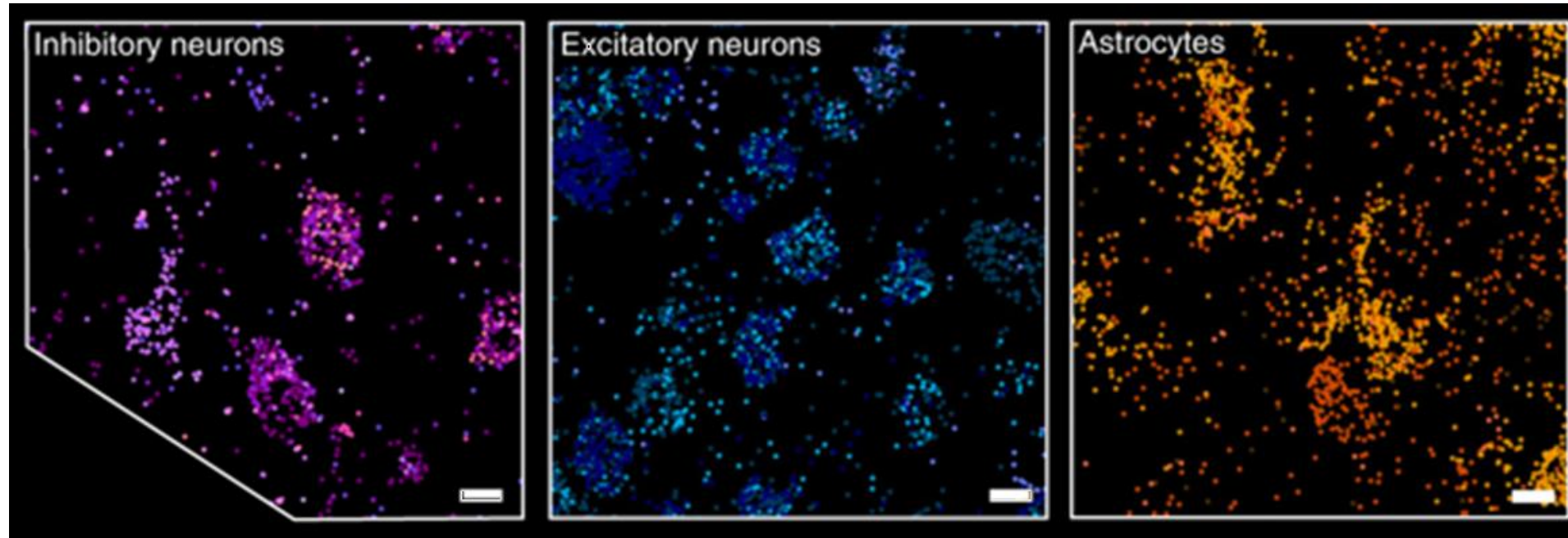
- | | Cells | Clusters | Semantic |
|----------------------------|---------------------|--------------------|-----------|
| 1 Anucleate red blood cell | one cell | one cluster | red |
| 2 Binucleate, no membrane | one cell over-split | one cluster | green |
| 3 Binucleate + membrane | one cell | one cluster | green |
| 4 Low density | nucleus + cytoplasm | correct | pink |
| 5 Dense, one type | merge | clusters ≈ 2 cells | one block |

Cell-segmentation-free analysis using SSAM and Sainsc

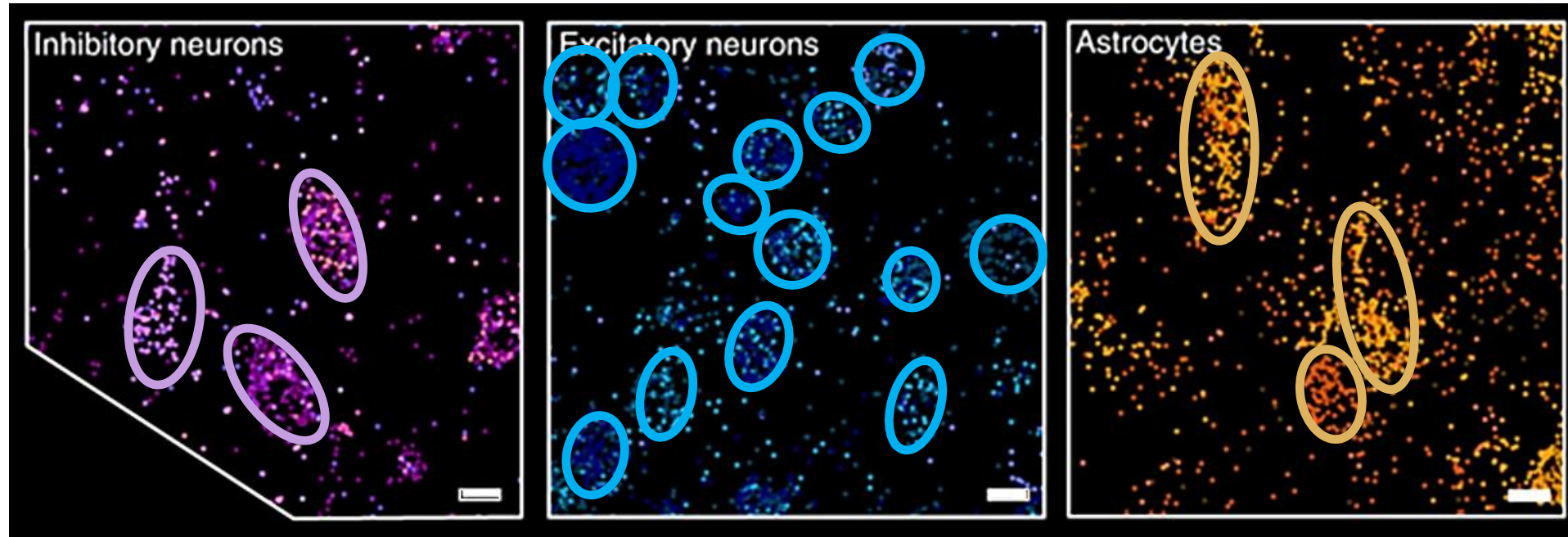
Park, Jeongbin et al. “Cell segmentation-free inference of cell types from in situ transcriptomics data.” Nature communications vol. 12,1 3545. 10 Jun. 2021, doi:10.1038/s41467-021-23807-4

Müller-Böttcher, Niklas et al. “Sainsc: A Computational Tool for Segmentation-Free Analysis of In Situ Capture Data.” Small methods, e2401123. 12 Nov. 2024, doi:10.1002/smt.202401123

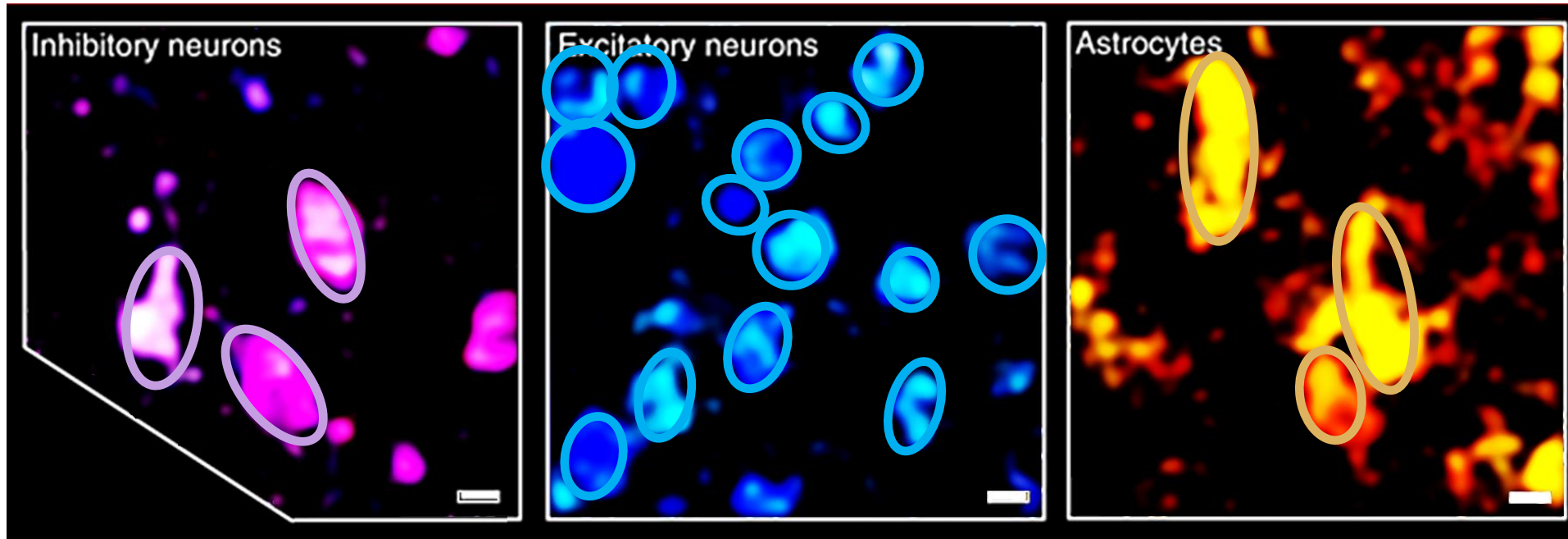
Transcript organisation patterns are not random



Transcript organisation patterns are not random

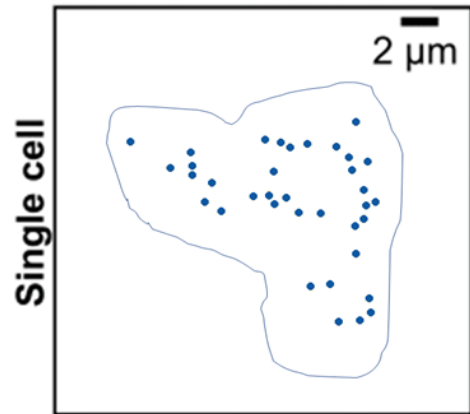


Transcript organisation patterns are not random



Modelling transcript density

SSAM and Sainsc model gene expression density



Modelling transcript density

SSAM and Sainsc model gene expression density



Modelling transcript density

SSAM and Sainsc model gene expression density



SSAM/Sainsc in 3 major steps

1. Smooth gene expression
2. Provide/identify cell type signatures
3. Generate the cell-type map (sematic segmentation)

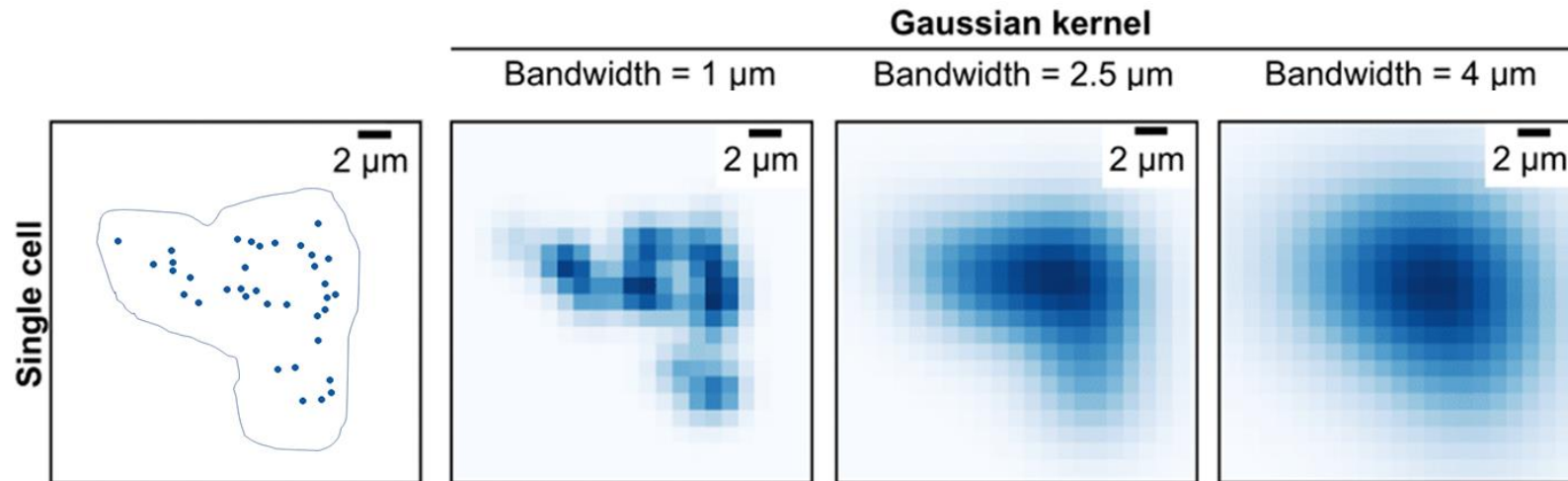
Step 1.1: calculate spatial mRNA density

Apply Kernel Density Estimation (KDE) with Gaussian kernel

Resulting image represents the probability density of mRNA existence

- *From discrete molecules to cloud of gene expression (“gene expression per pixel”)*

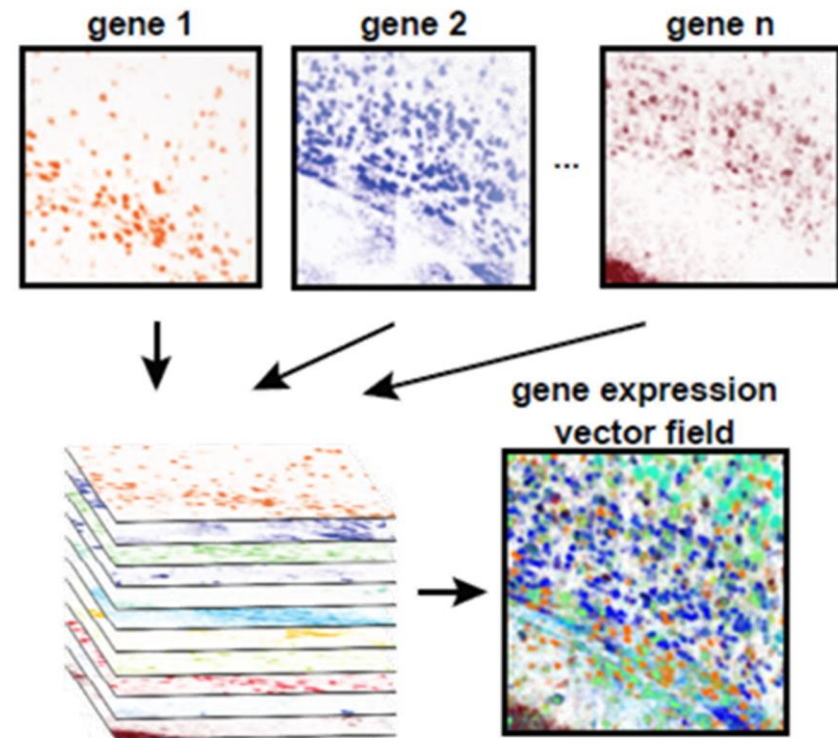
Bandwidth (sigma) should smooth between mRNA, but not outside of cells



Step 1.2: creating the “vector field” of gene expression

Stacking the KDE of each profiled gene creates the gene expression “vector field”

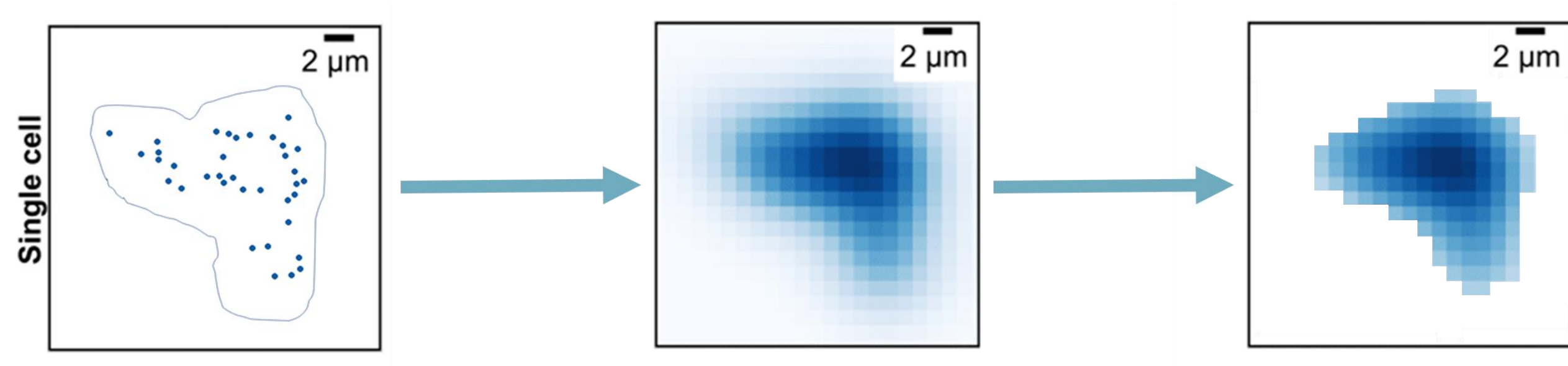
Each pixel in the vector field can be thought to have its own expression profile



Step 1.3: define gene expression threshold

If total gene expression density is too low then the signal likely originates from outside the cell

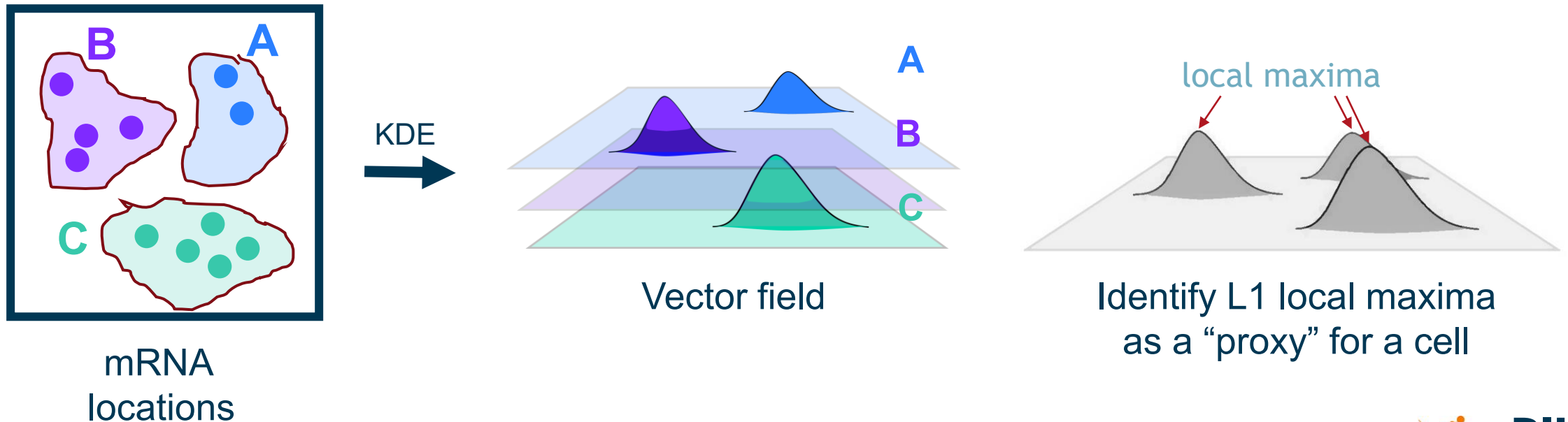
- Filtering low gene expression regions prevents classification of “low-quality” areas



Step 2.1: identify cell-type signatures

For many cell types, cell-type signatures are known: e.g. single-cell RNA sequencing

If cell-type signatures are not known then they need to be computed from the data
Selecting local maxima of mRNA signal as representatives of “cells”



Step 2.2: identify cell-type signatures - cluster local maxima

Cluster gene expression profiles of scRNAseq data or local L1 maxima

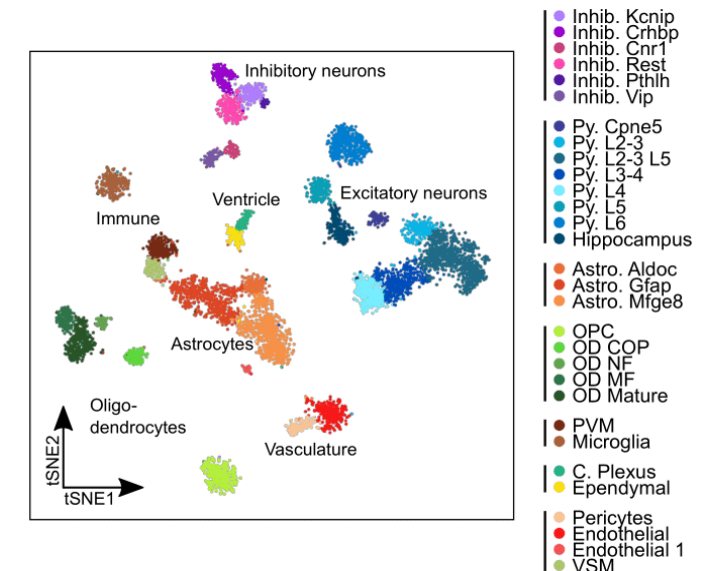
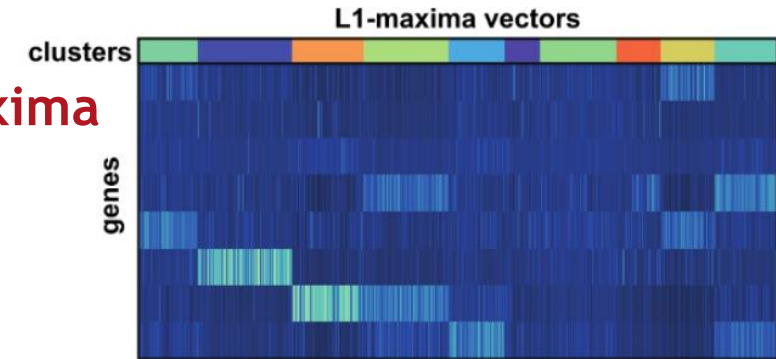
Different cluster = different cell type = different function!

SSAM adopts a Louvain algorithm clustering approach

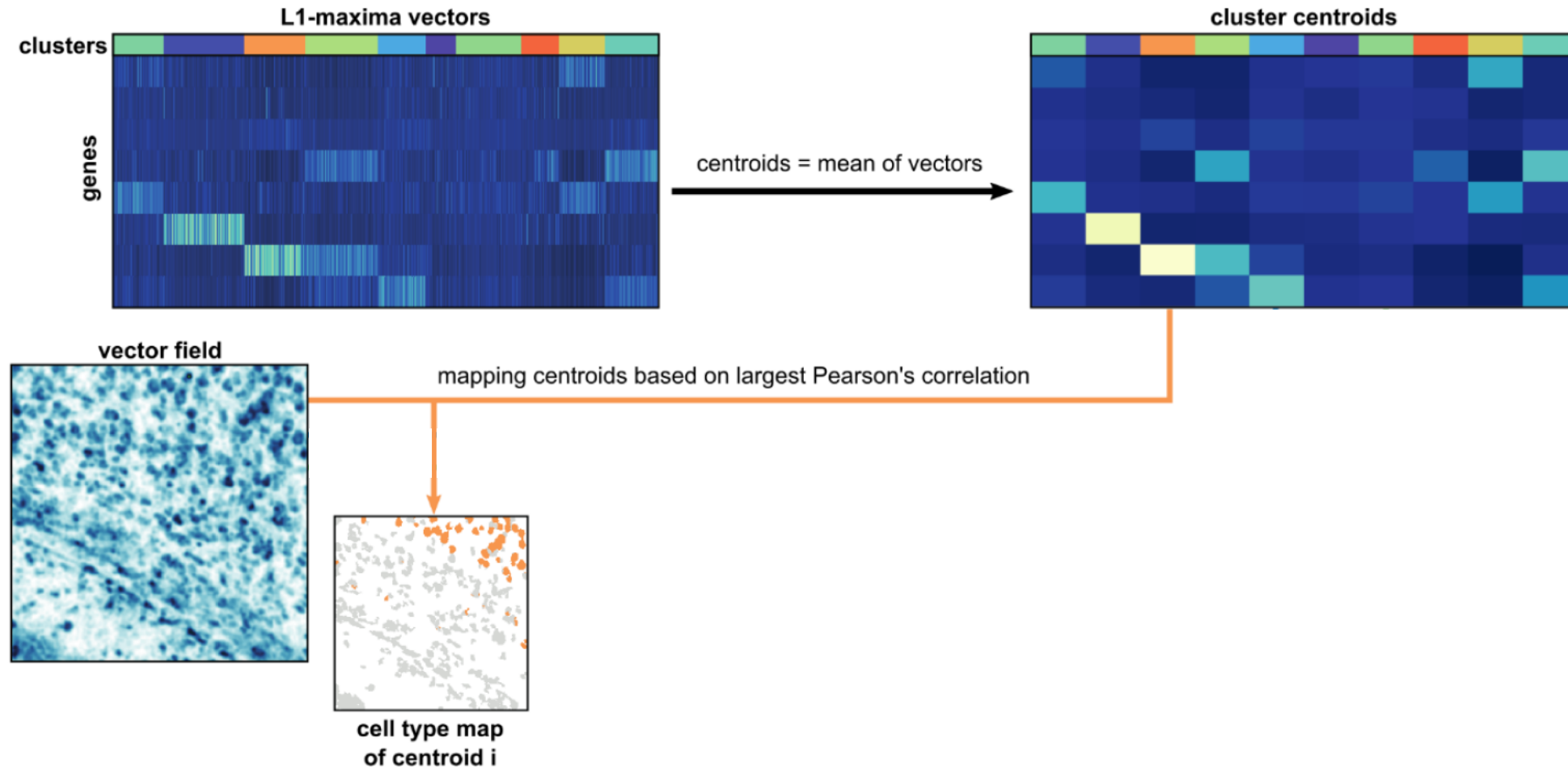
- This can be exchanged with your favourite clustering method!

Median cluster expression = cell-type gene expression signature

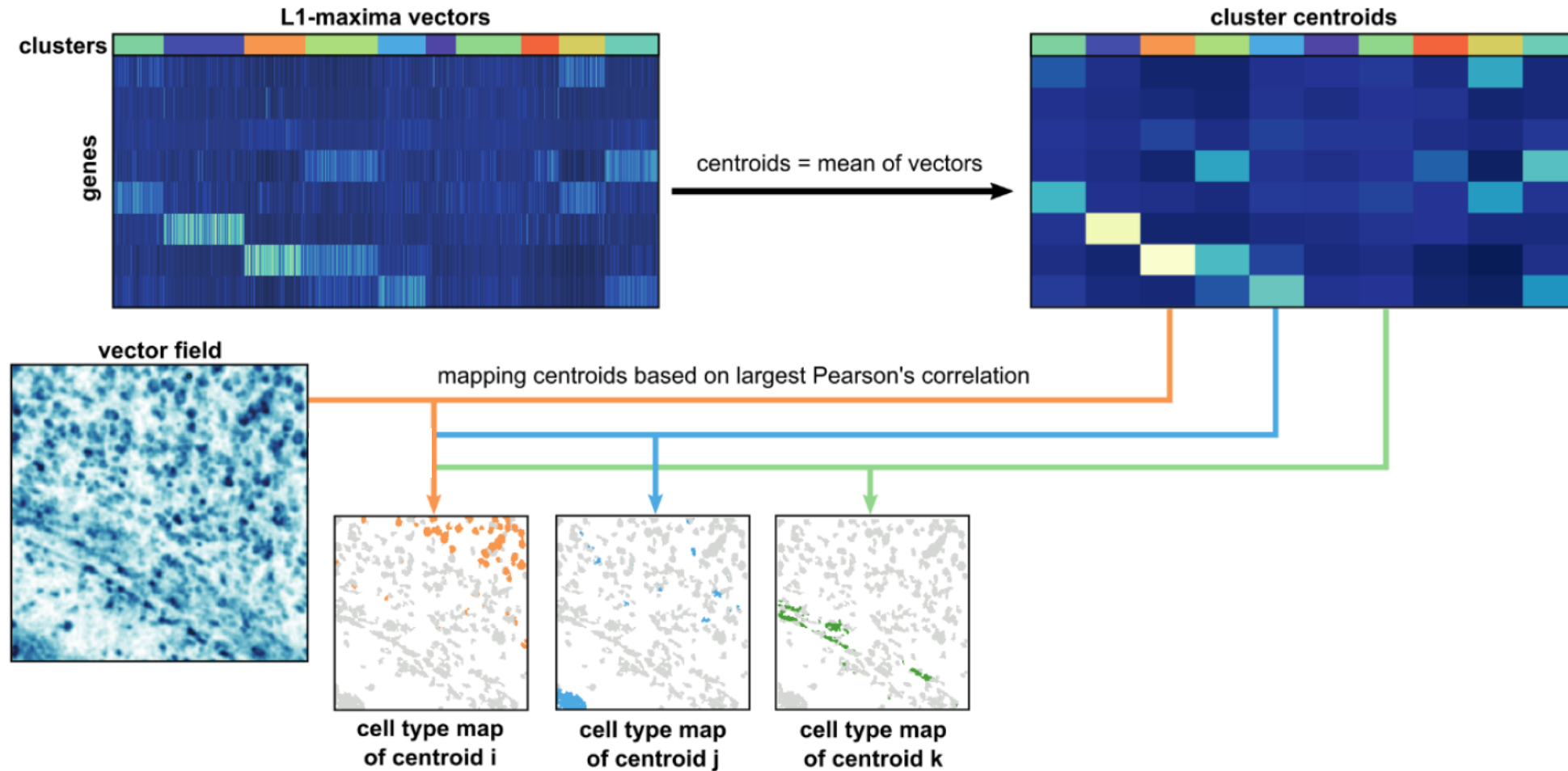
Visualise using PCA, tSNE, UMAP, ...



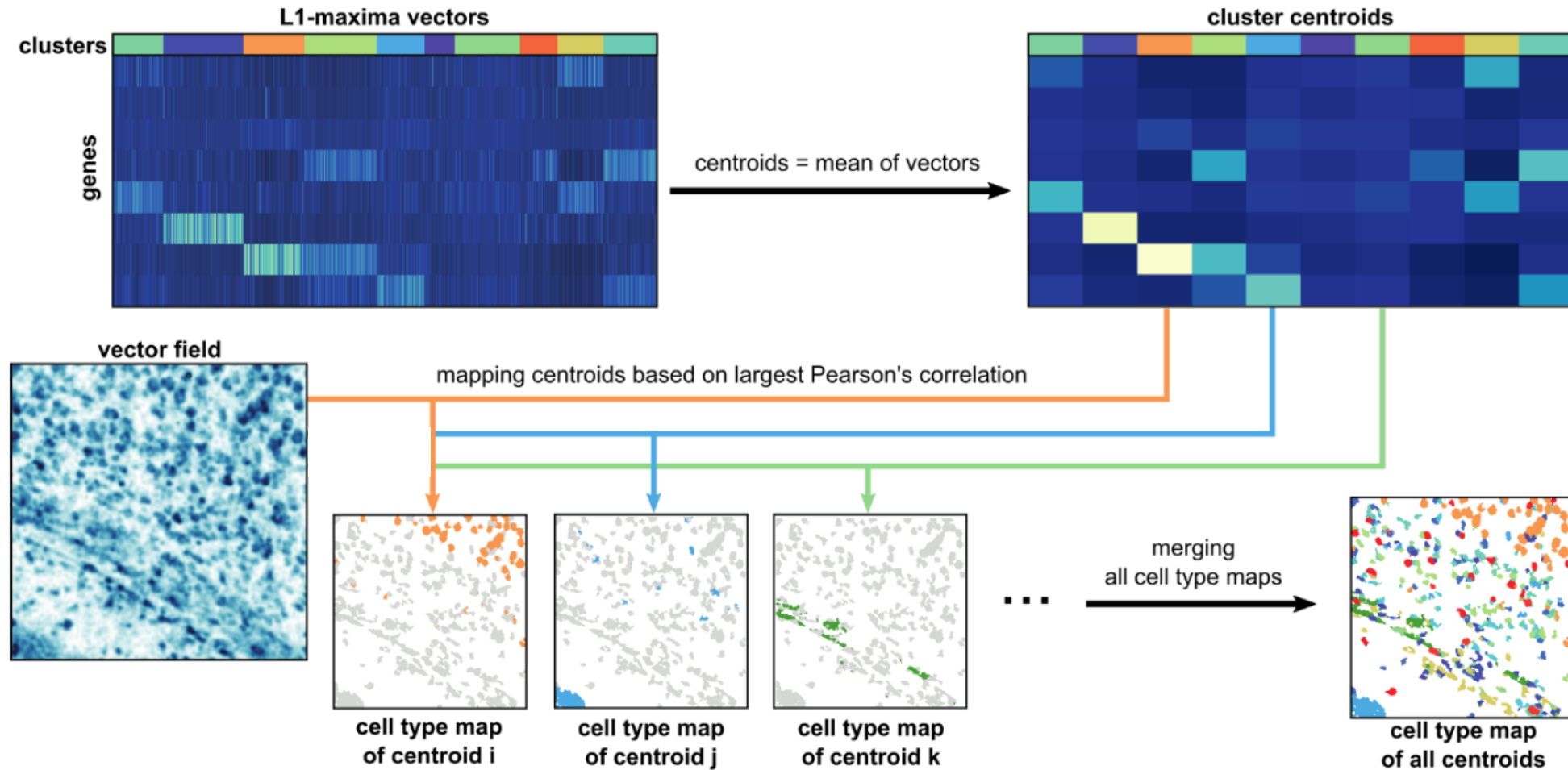
Step 3: generate the cell type map



Step 3: generate the cell type map



Step 3: generate the cell type map



Step 3: generate the cell type map

Classifying pixels

Pixels are classified based on a Pearson correlation

- Pixel gene expression VS cell-type gene expression signature
- Simple but effective
 - Works well when genes are robust cell type markers (i.e. low plex cell typing panels)
 - Doesn't work well when genes are not cell type specific (i.e. high plex gene panels)
- This step can be exchanged with your favourite ML classification method!

How does SSAM perform?

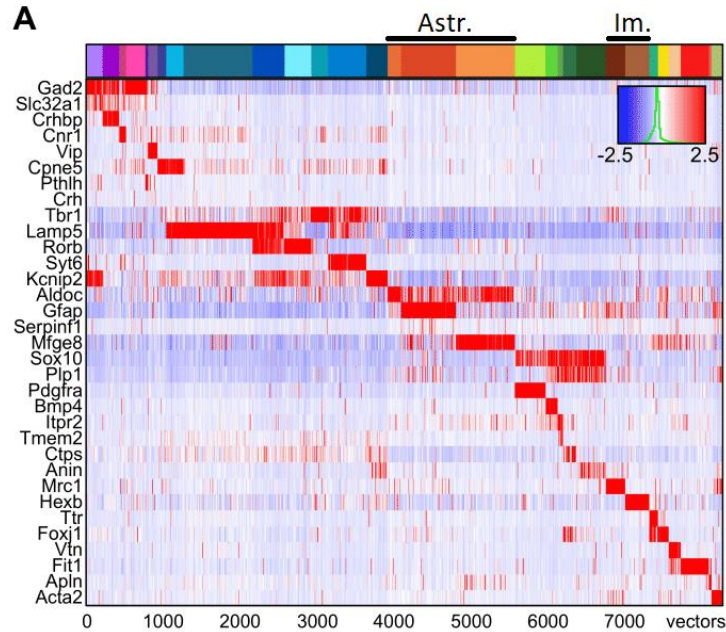
Adult mouse brain somatosensory cortex (SSp)

osmFISH, 35 genes

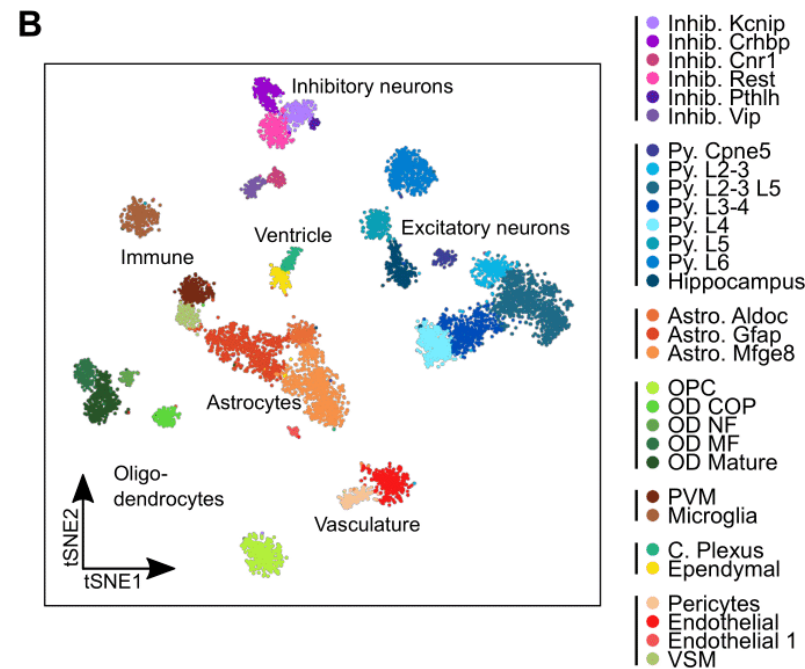
Codeluppi, Borm et al (2018), Nature Methods

SSAM identifies cell types accurately

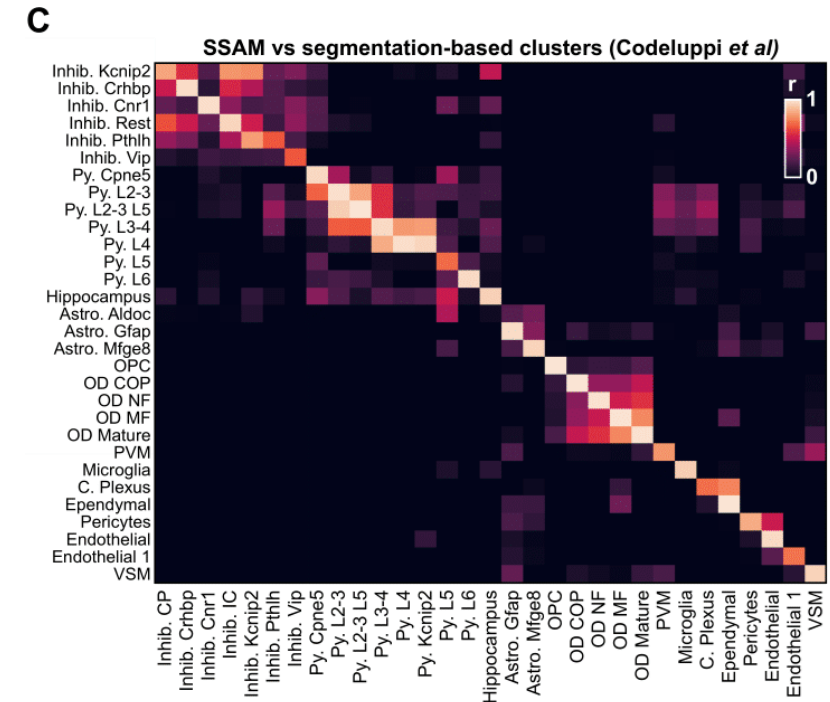
De novo cell-type signatures



Clustering

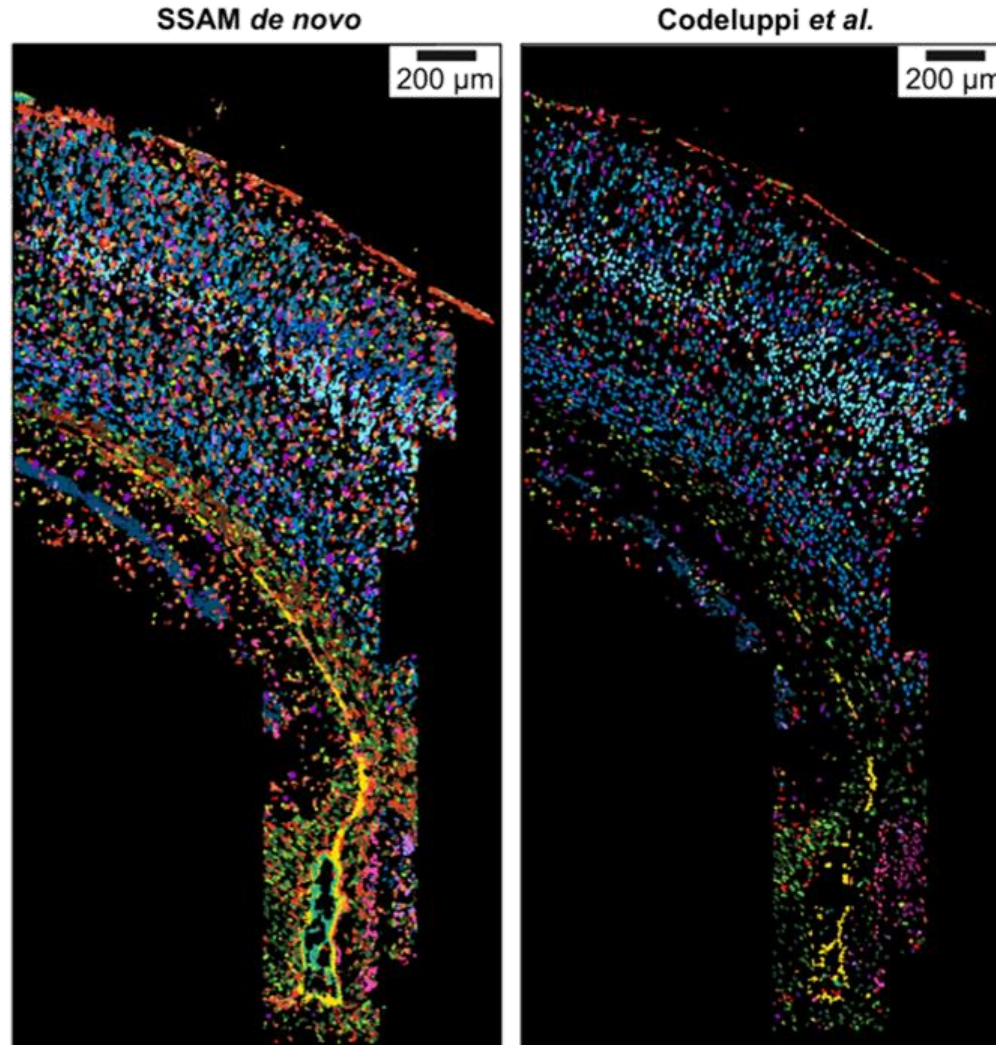


Signature comparison



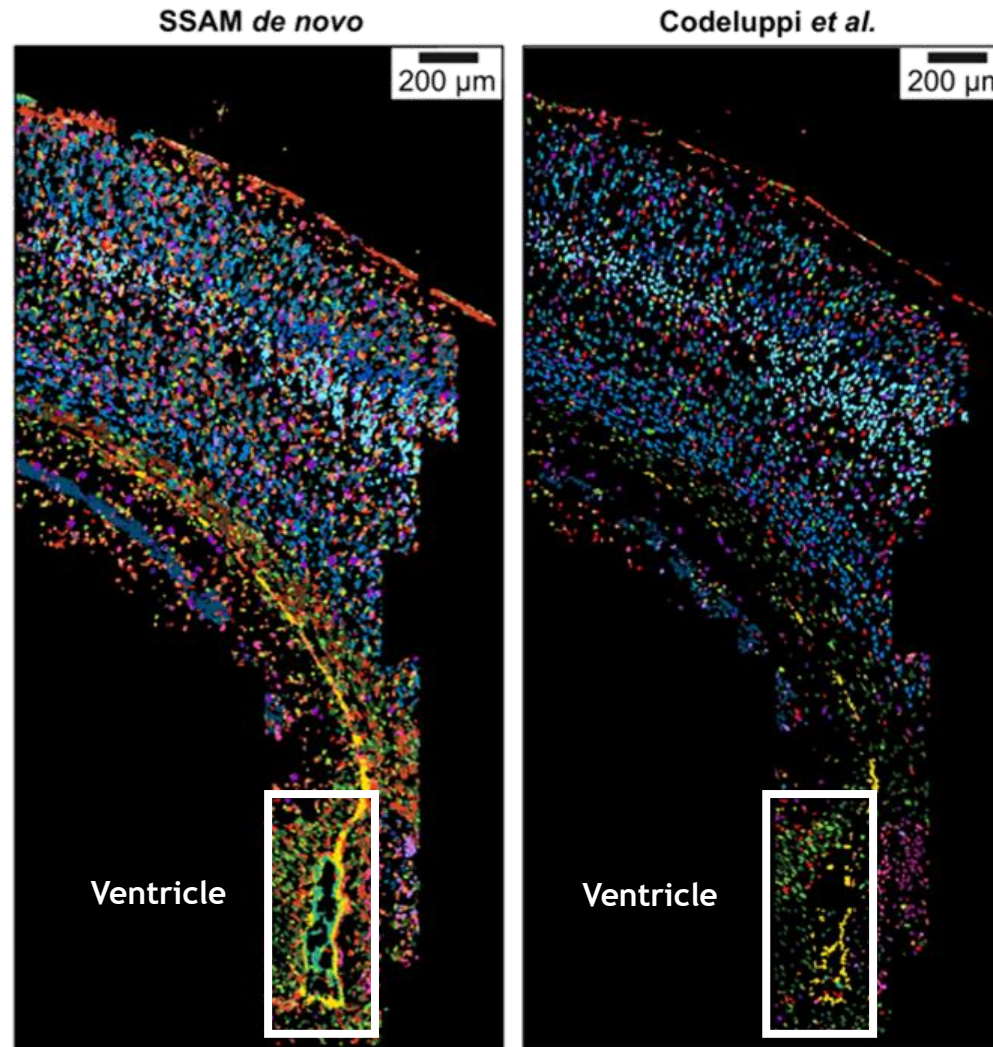
SSAM reconstructs the mouse SSp cell-type map

SSAM reconstructs mouse brain somatosensory cortex



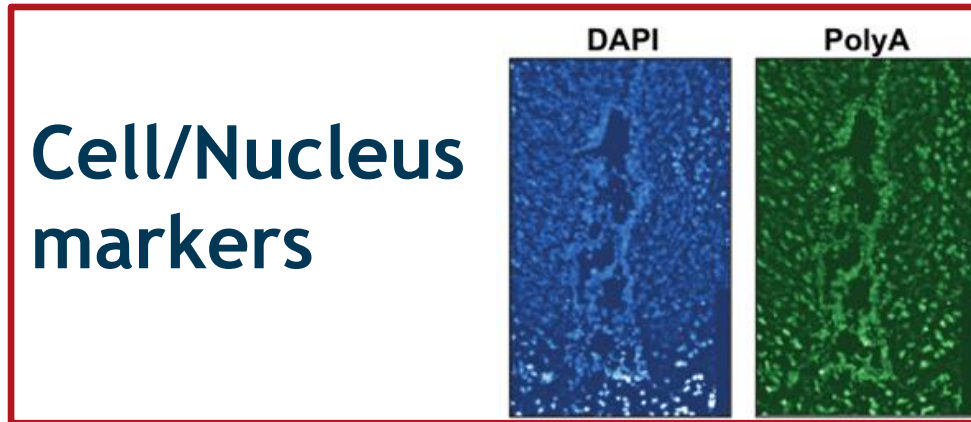
SSAM: how well does it work?

SSAM reconstructs mouse brain somatosensory cortex



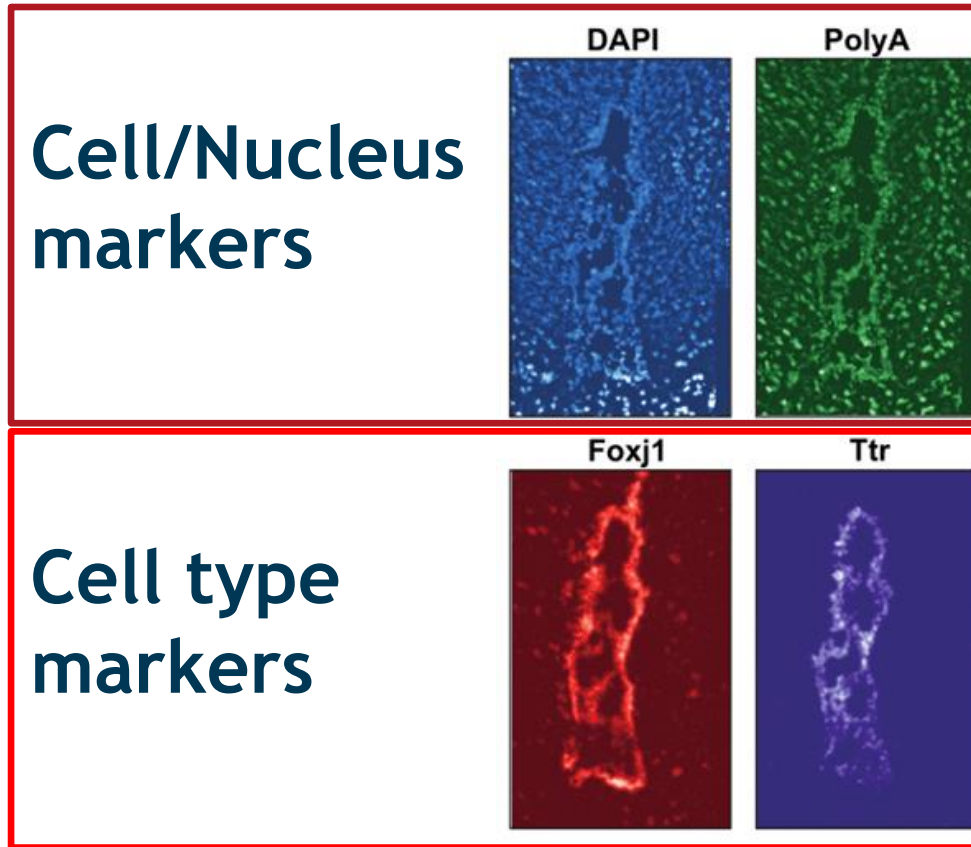
SSAM improves mapping of the ventricle region

Problem: low DAPI/Poly-A signal and occlusion



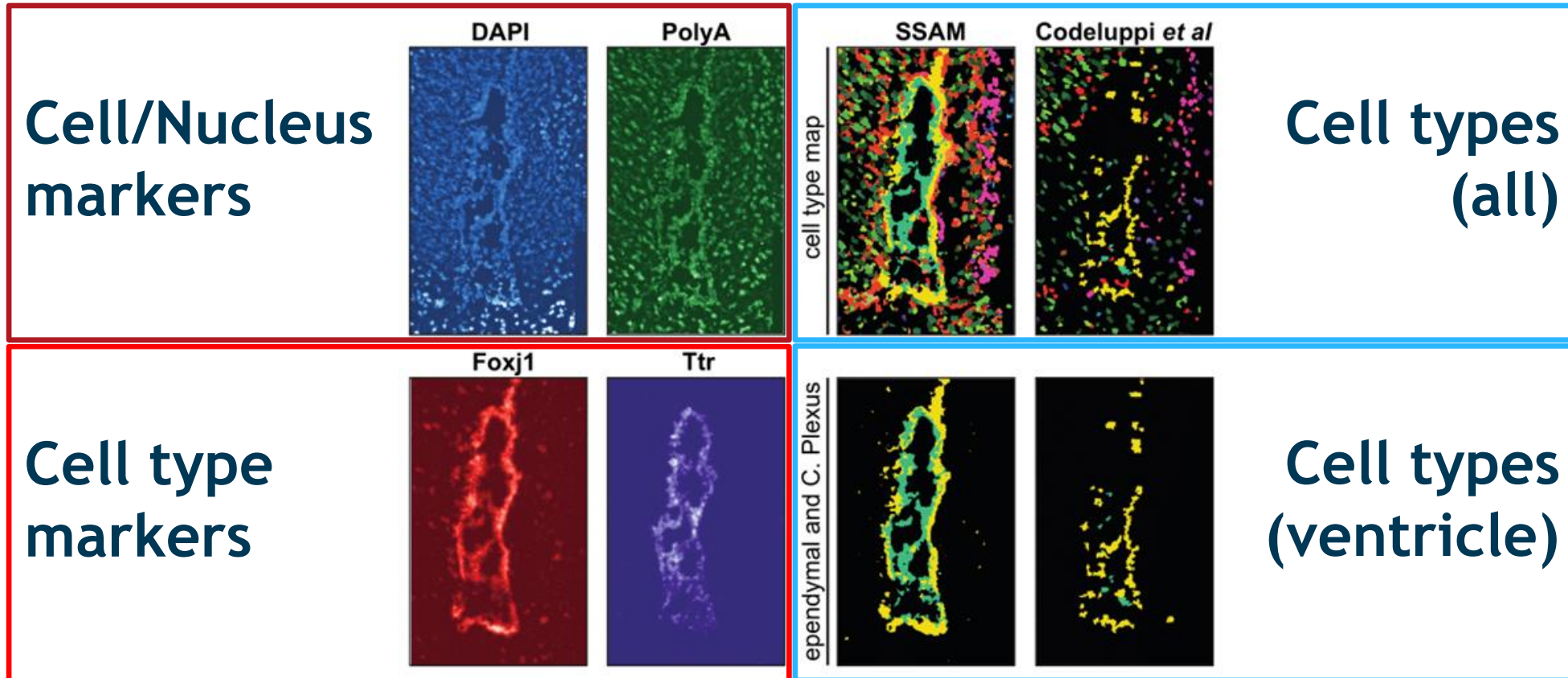
SSAM improves mapping of the ventricle region

Problem: low DAPI/Poly-A signal and occlusion, but high marker gene expression



SSAM improves mapping of the ventricle region

Problem: low DAPI/Poly-A signal and occlusion, but high marker gene expression



Emerging state of the art for spatial transcriptomics

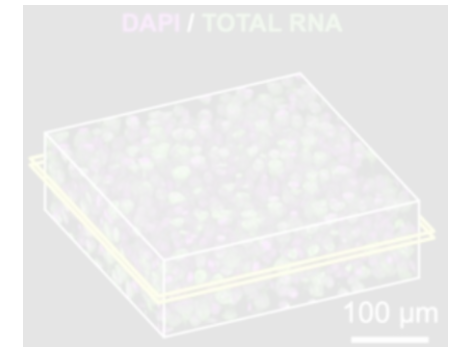
Size and resolution



Pseudo 3D

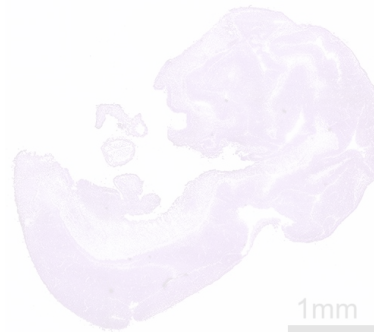


Real 3D



Multi-omics

H&E (E.10 Embryo)



Pan-mRNA map



Pan-protein map



Sainsc: optimising SSAM for millions of cells & organism-scale

Segmentation-free Analysis of IN Situ Capture data

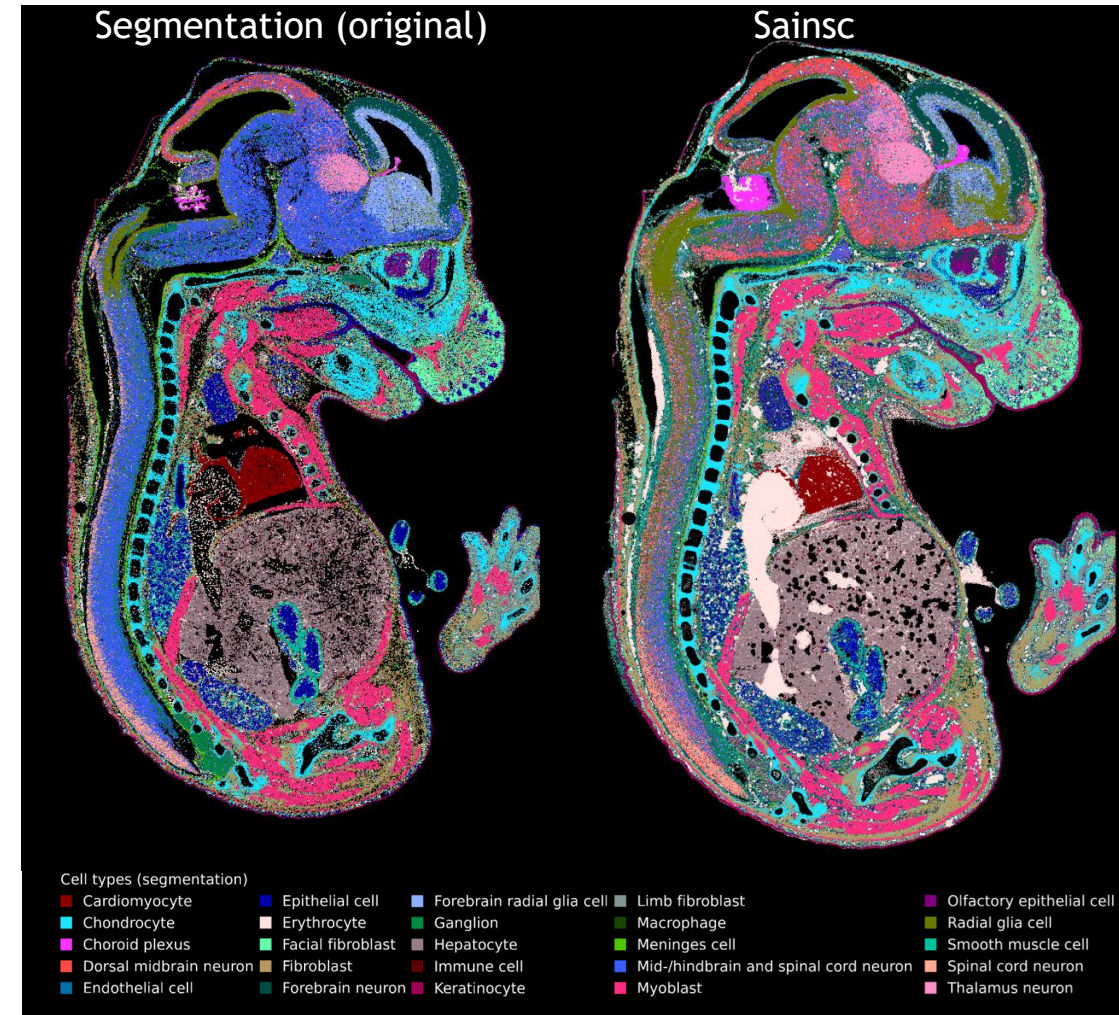
- Segmentation-free identifies red blood cells in the spleen and umbilical cord

Optimisation for organism scale analysis

- 10,000 times faster than SSAM
- 100 times less memory usage than SSAM

Suitable for imaging and sequencing-based spatial transcriptomics

- E.g. Stereo-seq, Open-ST, Nova-ST, VisiumHD



Sainsc: optimising SSAM for millions of cells & organism-scale

Segmentation-free Analysis of IN Situ Capture data

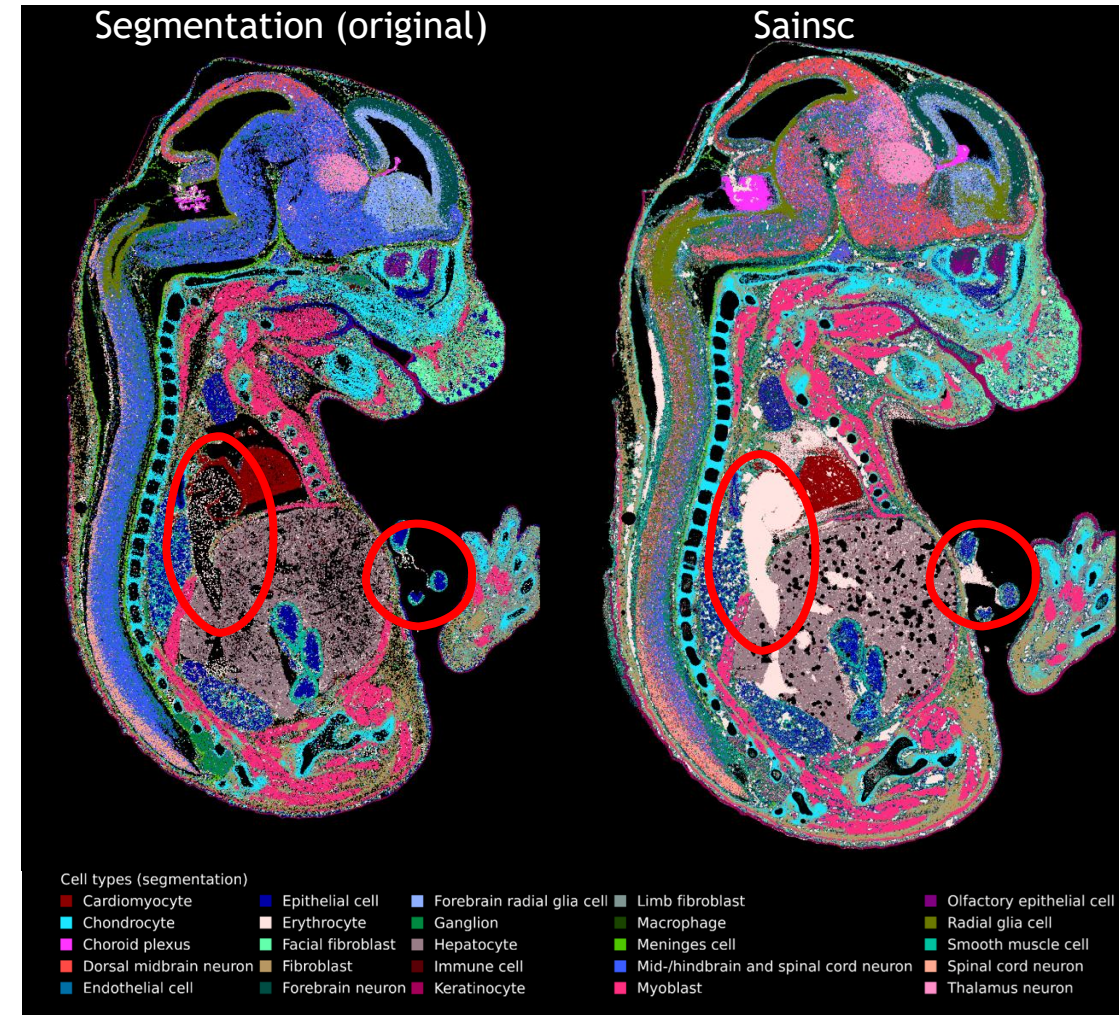
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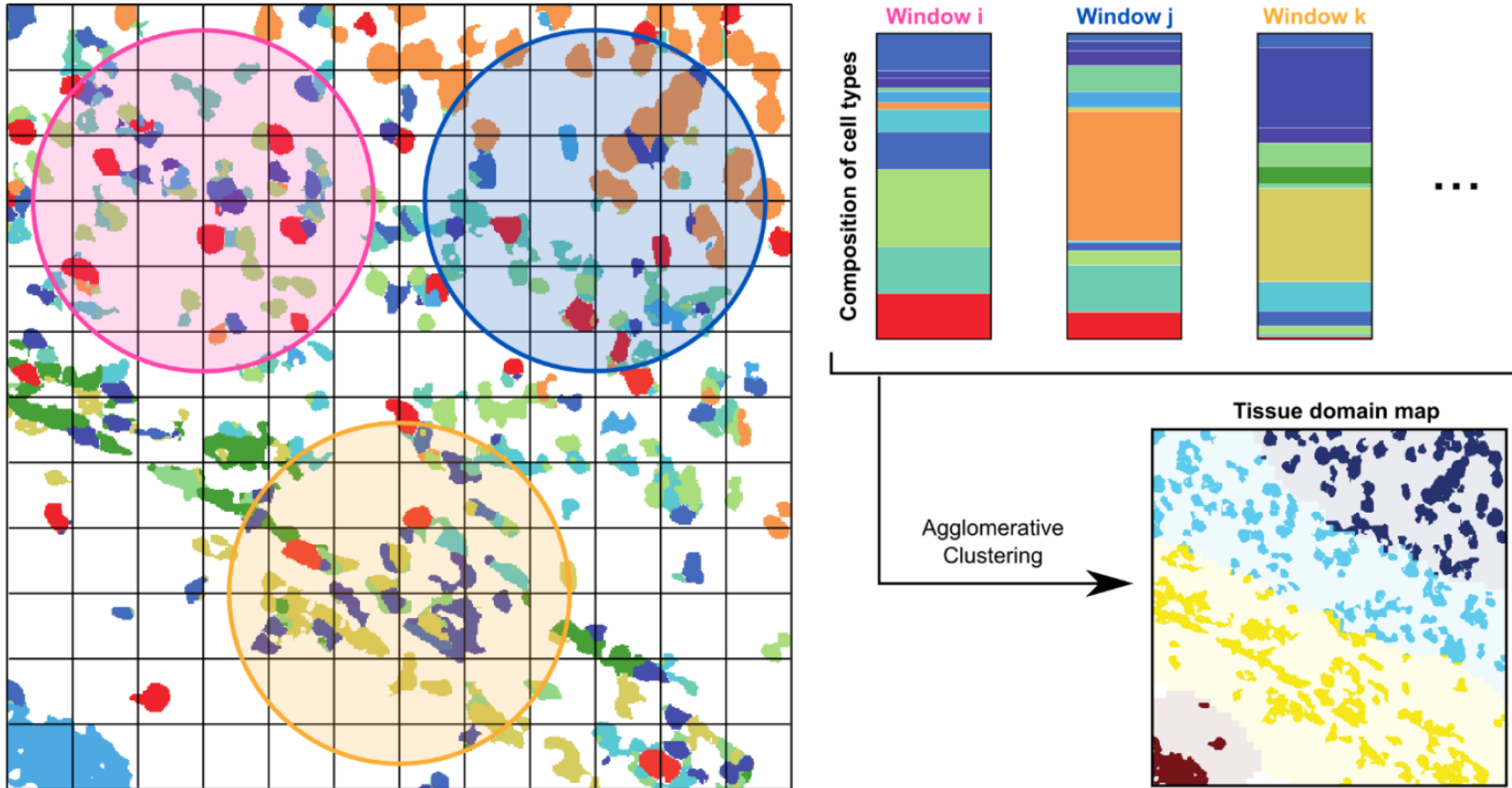
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- E.g. Stereo-seq, Open-ST, Nova-ST, VisiumHD

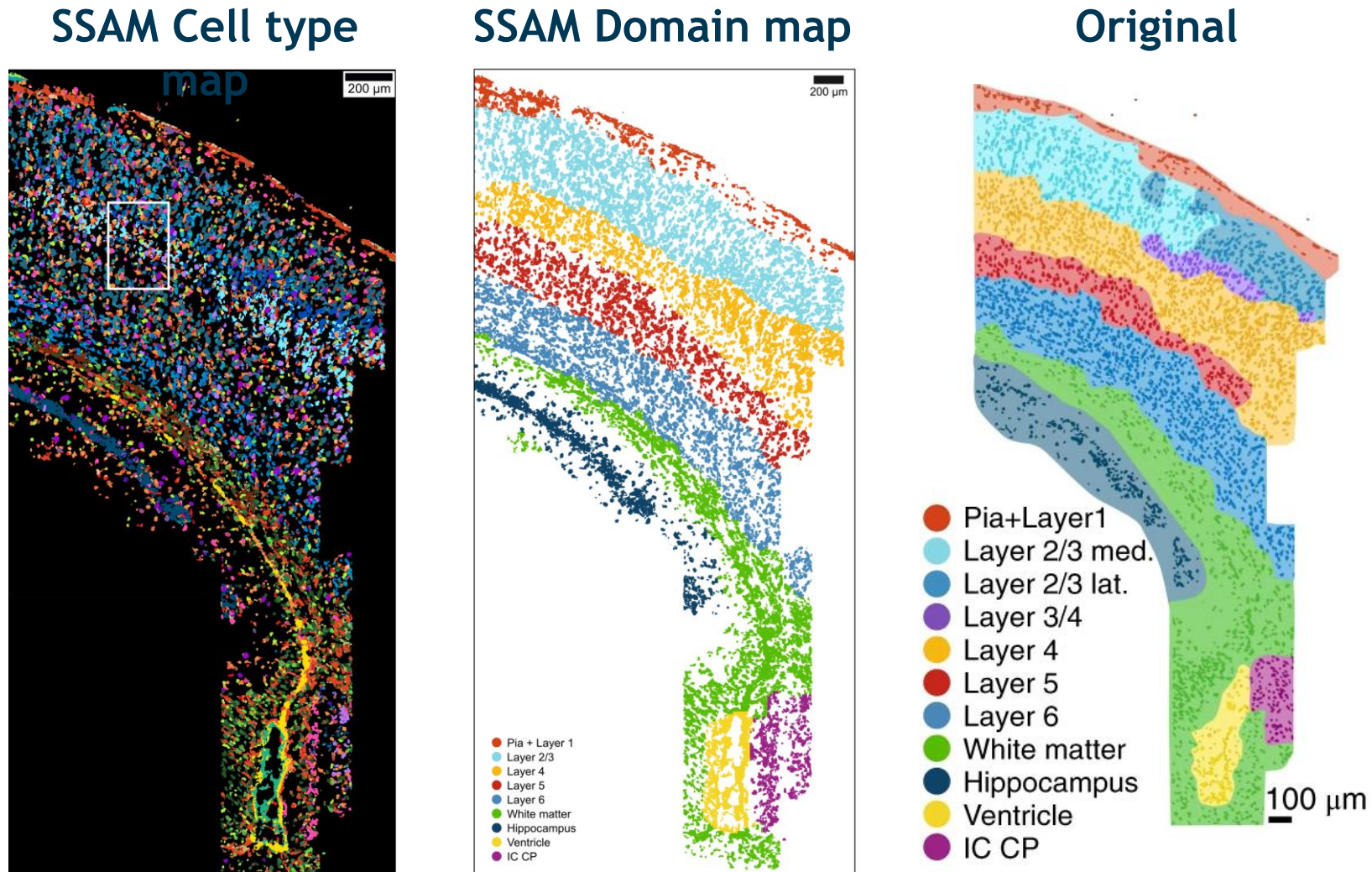


But what can you do without cells?

Downstream analysis - spatial domains



SSAM identified mouse SSp cortical layers



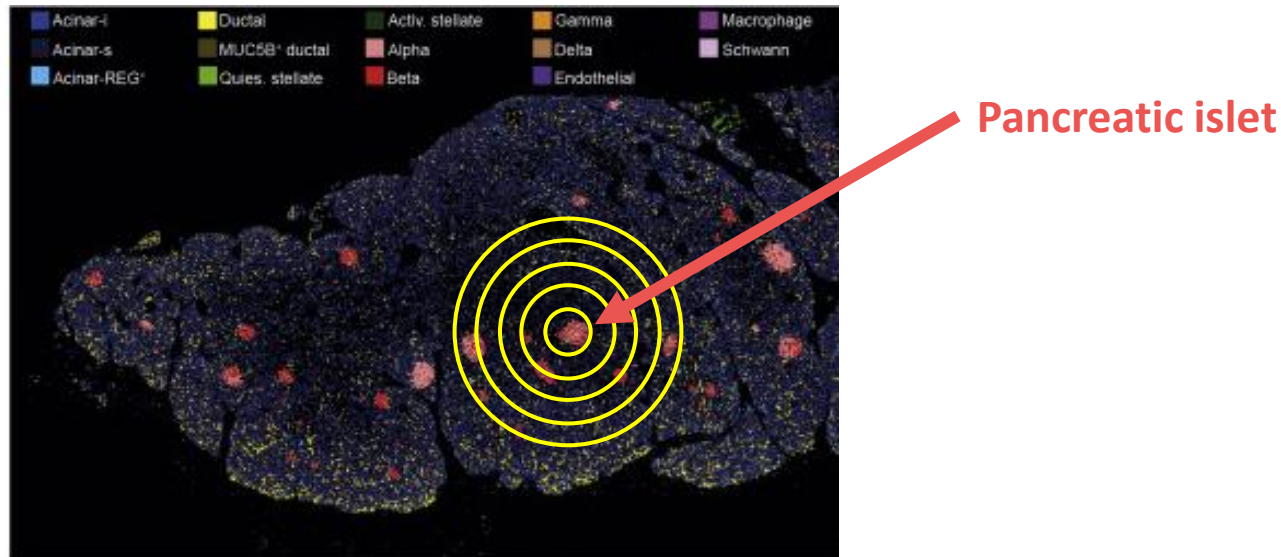
CellSonar: spatial relationships between cell types

Statistical modelling of spatial relationships in the pancreas



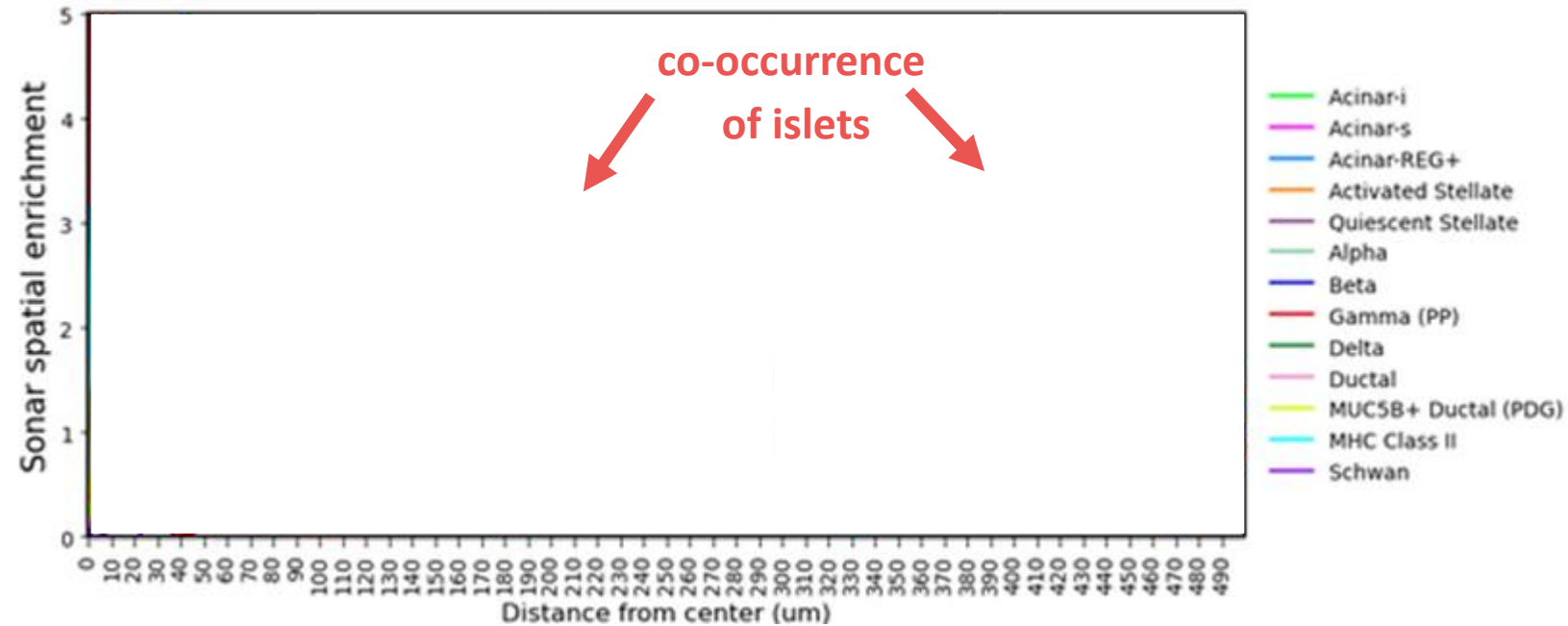
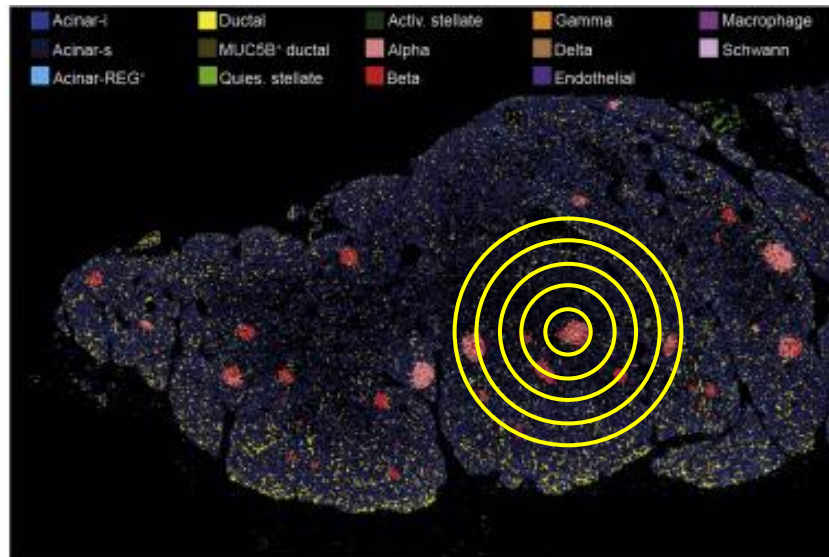
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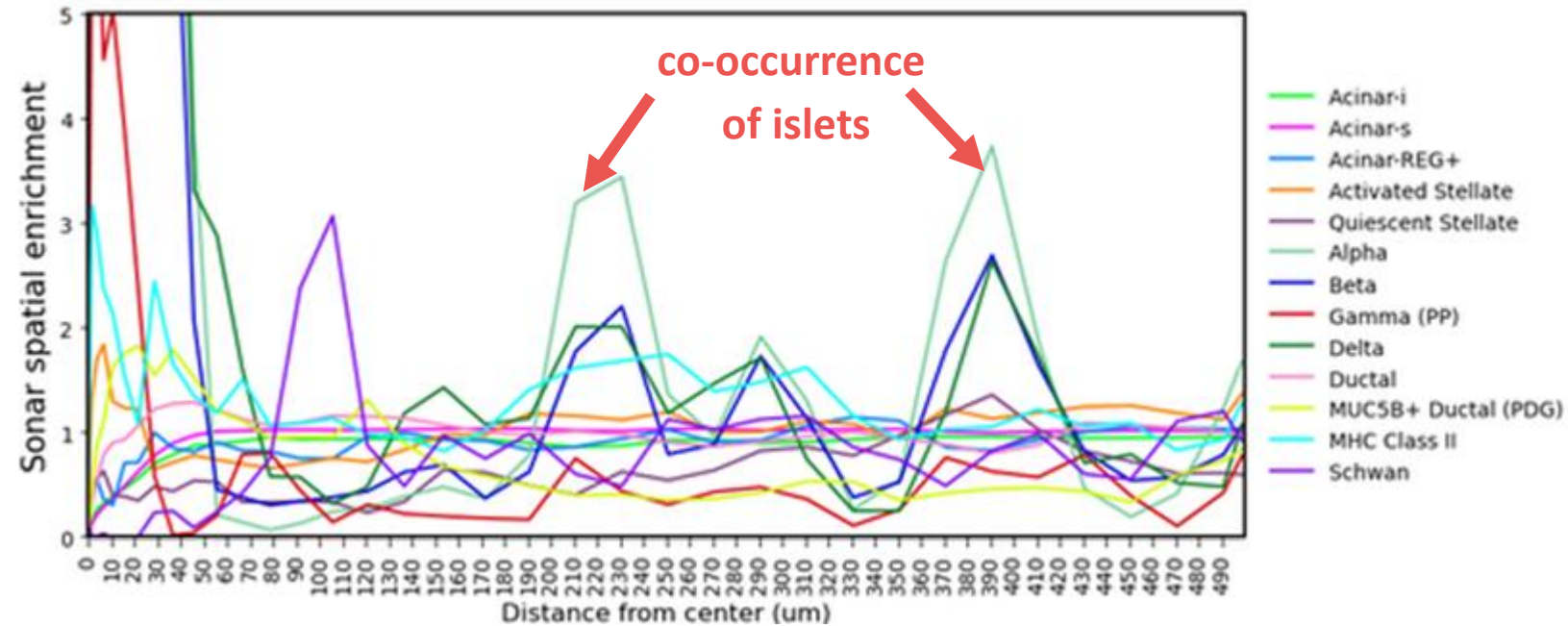
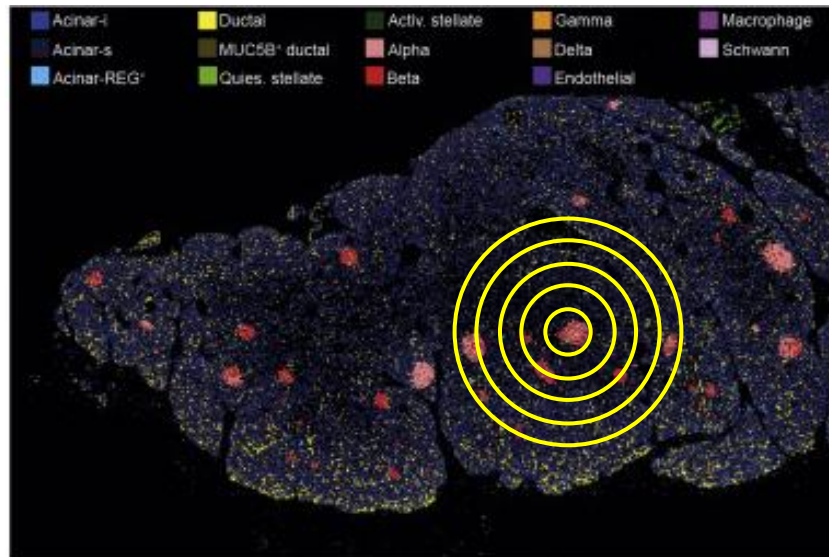
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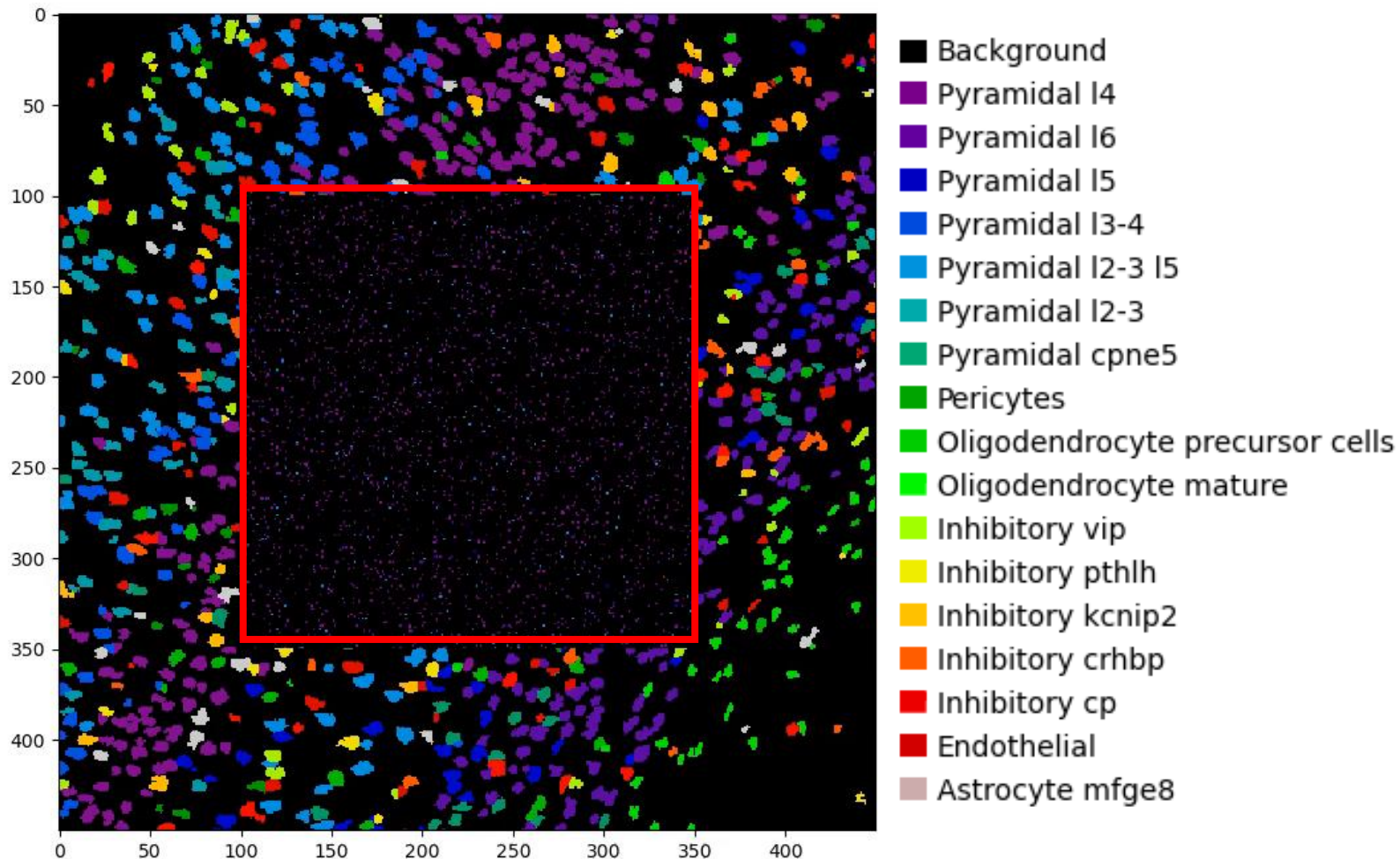


CellSonar: spatial relationships between cell types

Statistical modelling of spatial relationships in the pancreas



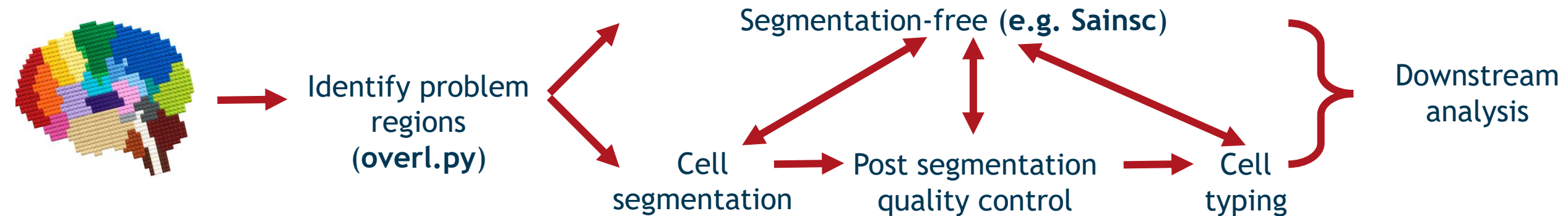
CellSonar: generative capabilities (click play!)



Summary

- Bad cell segmentation can lead to inaccurate analysis and missing important signal
- Cell-segmentation-free approaches are powerful parallel analysis avenue
- Transcript organisation can be used to infer cellular architecture of tissues
- Cell-segmentation-free differs from cell segmentation:
 - (i) semantic segmentation (e.g. a map of pixels classified by cell type)
 - (ii) instance segmentation not of cells (e.g. cell parts)

Proposed workflow



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SpaceHack Participants

HCA Cell Segmentation Team

AG-Ishaque



Postdoc



PhD



Clinician Scientist



Swiss Institute of Bioinformatics



Swiss National Science Foundation



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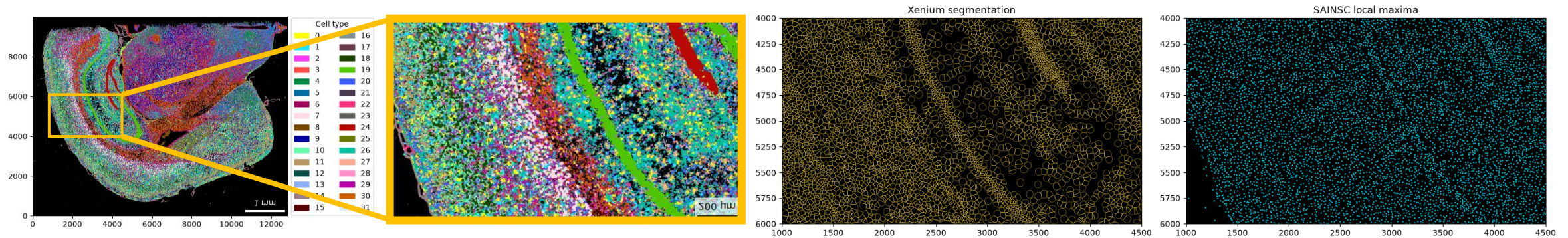
Universität Zürich



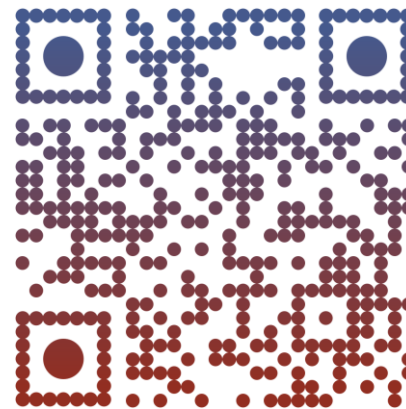
Practical P3: cell-segmentation-free analysis using Saisnc

https://github.com/elixir-europe-training/ELIXIR-SCO-spatial-omics-2026/blob/main/practicals/practical_3/p3_segmentation_saisnc.ipynb

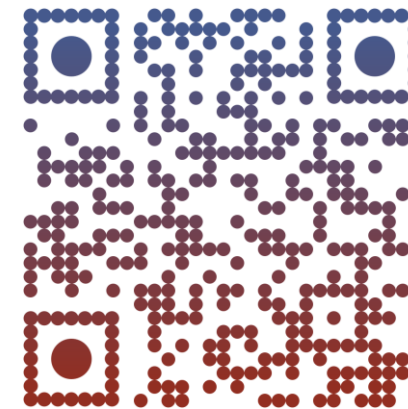
- Using the Saisnc tool to perform cell-segmentation-free analysis on Xenium mouse brain
- Trainers: Dr. Aditya Singh (NBIS, SE) and Dr. Lucie Pfeiferova (CAS, CZ)



Tutorial



Saisnc GitHub



Saisnc Docu